

Methods in Nucleic Acid Sequencing

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Johns Hopkins University

NGS instruments and applications

History of sequencing instruments

- Successful approaches
- Failed approaches (lessons to be learned)

NGS approaches – how did Illumina come to dominate?

Applications & Requirements

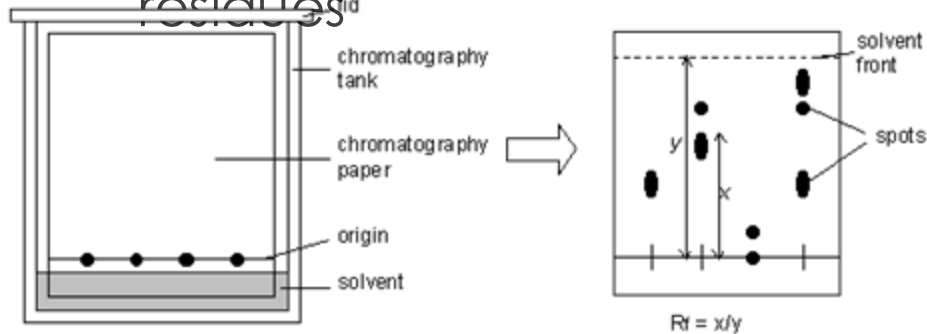
1st Generation Sequencing Era

Sanger - 2 Nobel prizes for Chemistry
Protein Seq (1951 Sanger)

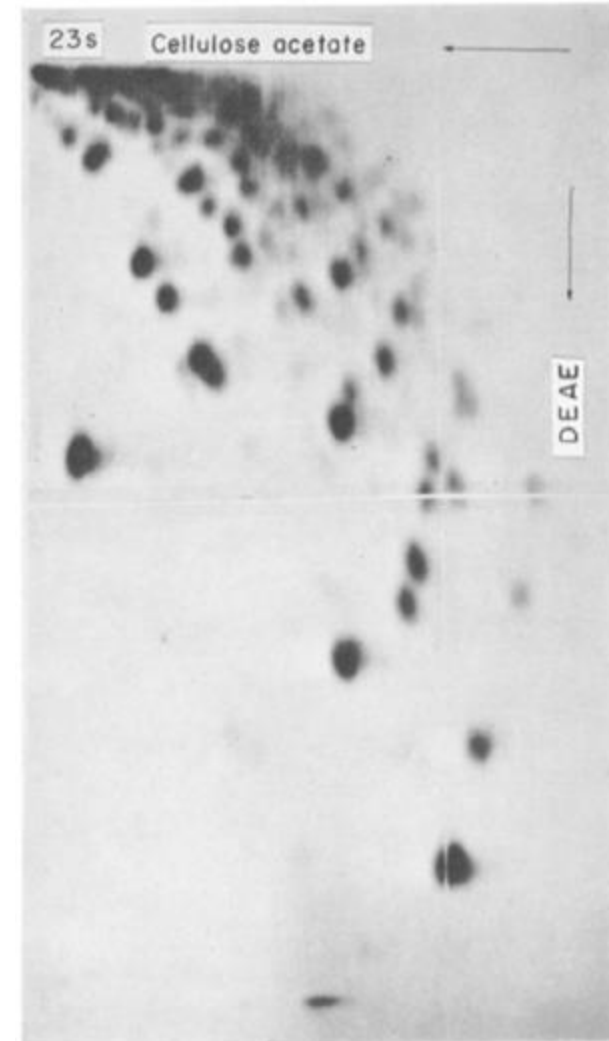
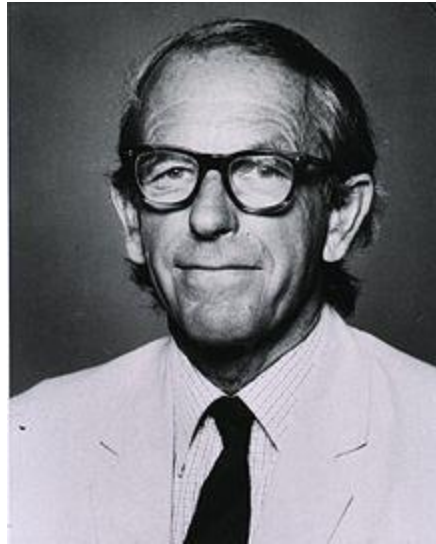
- Sequencing via 2-D chromatography

RNA seq (1965 Sanger)

- Radiolabelled RNA residues



<http://www.biologymad.com/master.html?http://www.biologymad.com/biochemistry/biochemistry.htm>



Sanger, et. al. *J. Mol. Biol.* 1965

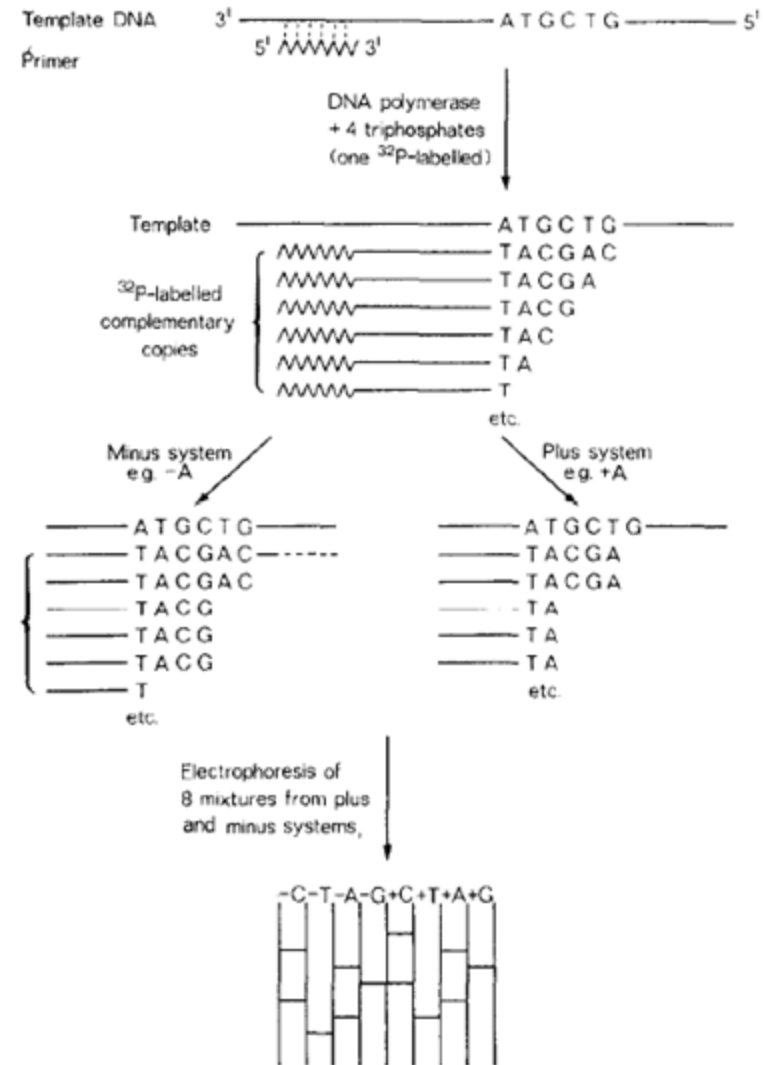
1st Generation Sequencing Era

A Rapid Method for Determining Sequences in DNA by Primed Synthesis with DNA Polymerase

F. SANGER AND A. R. COULSON

J. Mol. Biol. (1975) **94**, 441–448

- “Plus-Minus” sequencing was the first approach to practical sequencing.
- The “Minus” system involves polymerase extension with only 3 dNTPs (i.e. CTP, GTP, TTP). Extension arrests at locations of the missing dA.
- The “Plus” system involves digesting strands with exonuclease in the presence of polymerase and only one dNTP (i.e. dATP) - digestion arrests at locations of the dATP.
- Separation on size then gives locations and length of A sequences.



1st Generation Sequencing Era

A new method for sequencing DNA

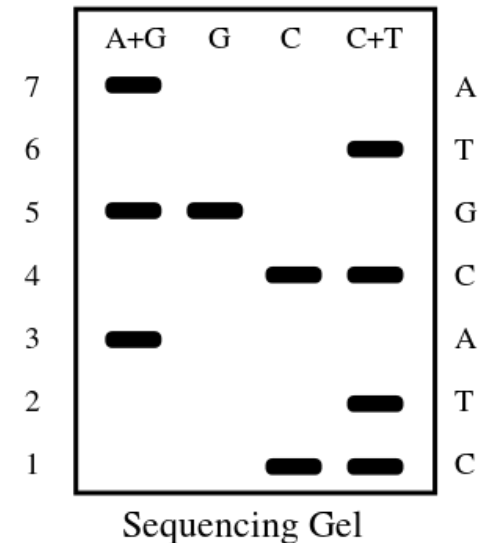
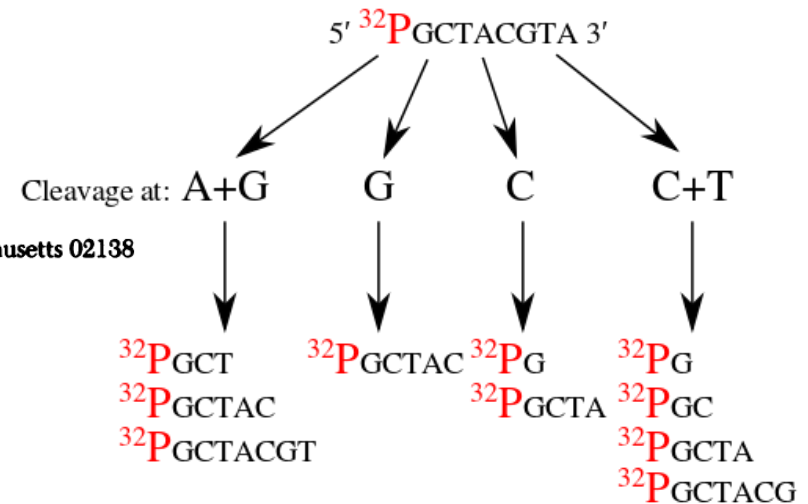
(DNA chemistry/dimethyl sulfate cleavage/hydrazine/piperidine)

ALLAN M. MAXAM AND WALTER GILBERT

Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts 02138

Contributed by Walter Gilbert, December 9, 1976

- Maxam-Gilbert DNA sequencing relies on selective chemical cleavage of DNA, followed by a piecing back together of the sequence
- As in Plus-Minus sequencing, the strands are labeled on the 5' (or leading edge) and separated based on size.



Wikipedia

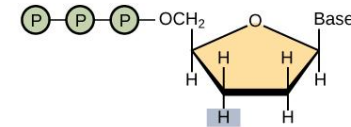
1st Generation Sequencing Era

DNA sequencing with chain-terminating inhibitors

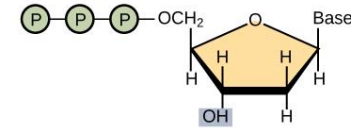
(DNA polymerase/nucleotide sequences/bacteriophage ϕ X174)

F. SANGER, S. NICKLEN, AND A. R. COULSON

- Chain terminator method by Sanger/Coulson
- Uses doped in di-deoxynucleotides (ddNTPs), terminating and blocking synthesis a random fraction of the time.
- Priming strands are radioactively labeled, extended with a mix of ddNTP and dNTPs, then size separated.

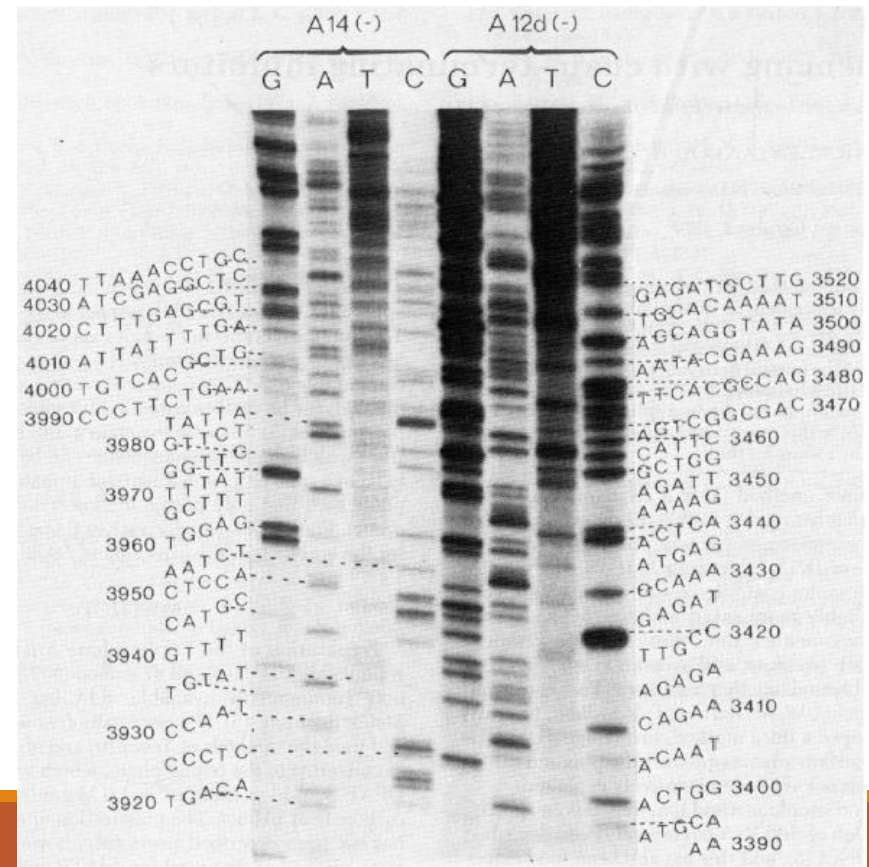


Dideoxynucleotide (ddNTP)



Deoxynucleotide (dNTP)

OpenStax Biology

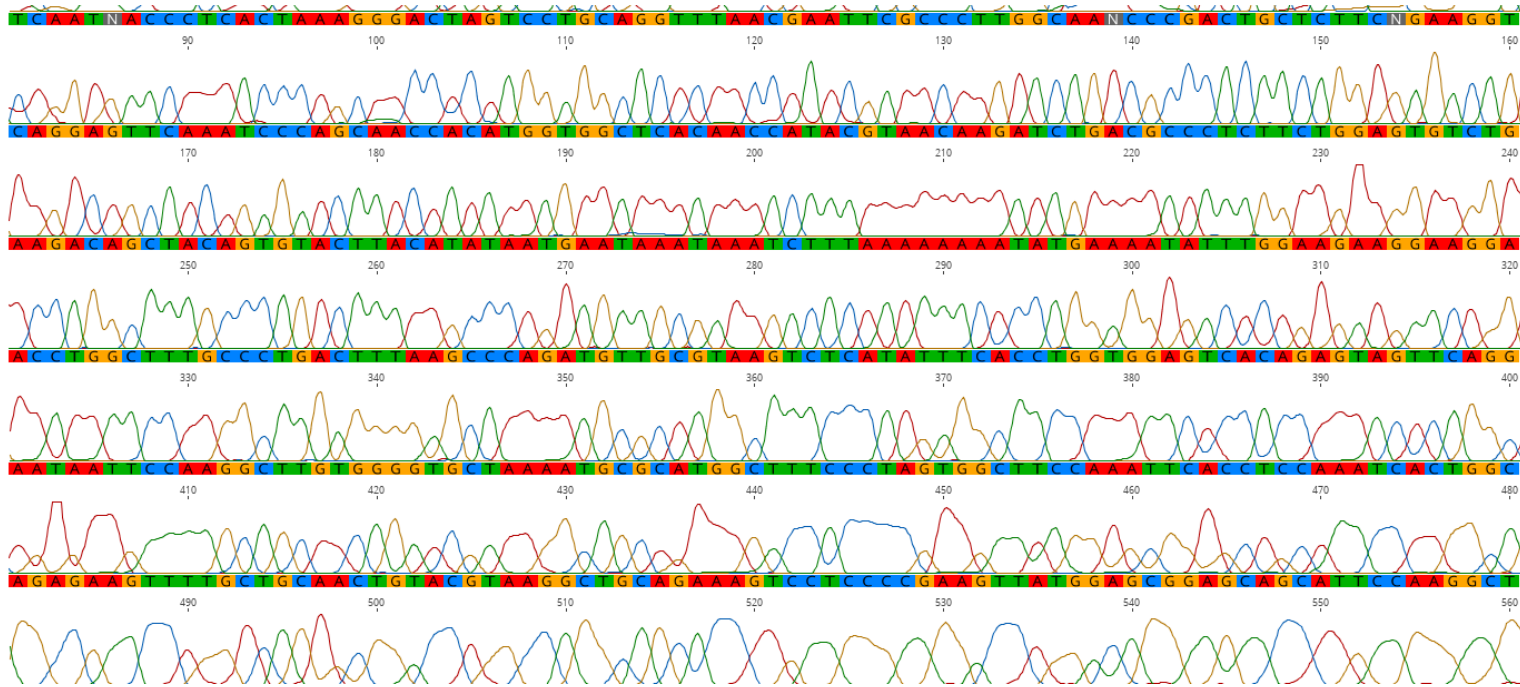


1st Generation Sequencing Era

NATURE VOL. 321 12 JUNE 1986

Fluorescence detection in automated DNA sequence analysis

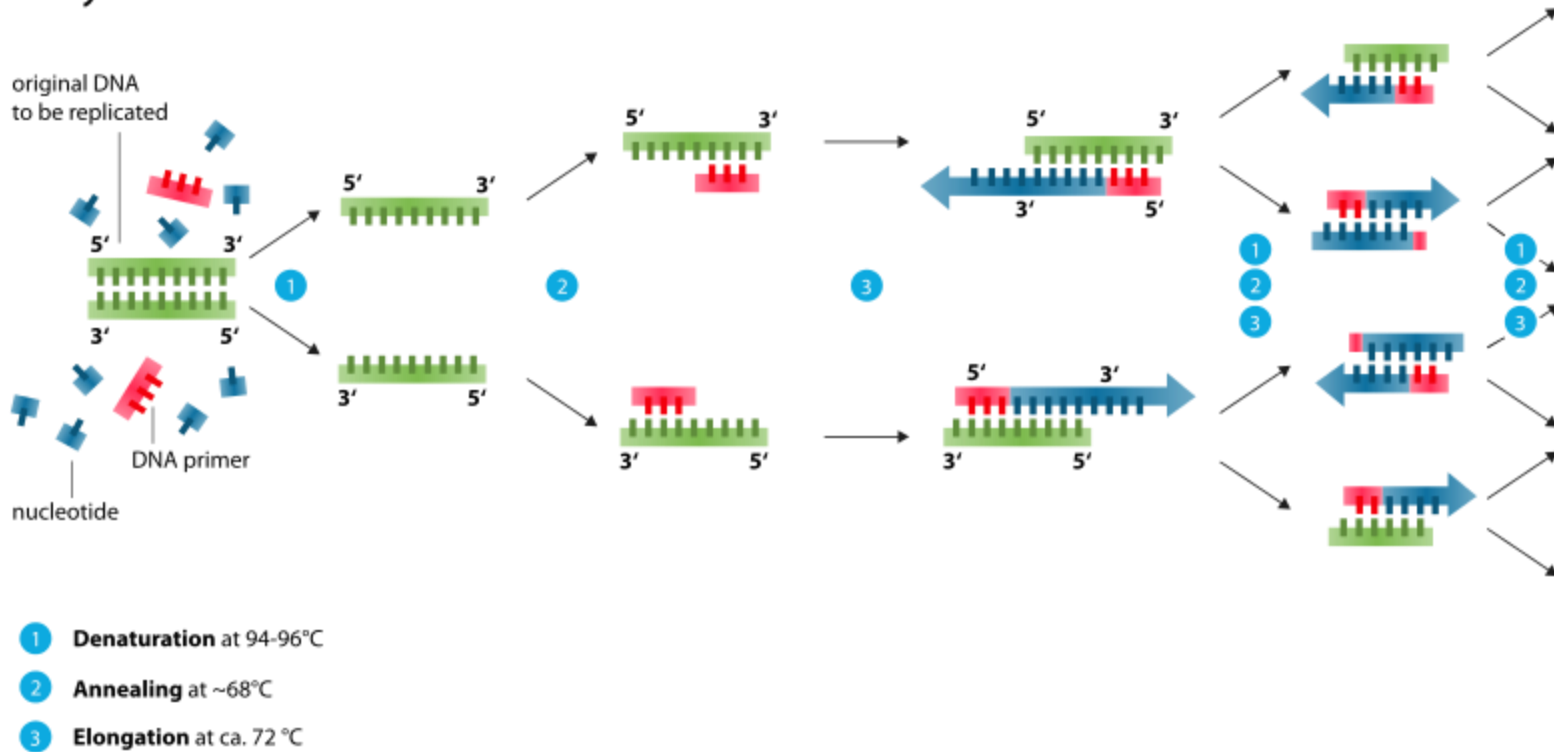
**Lloyd M. Smith, Jane Z. Sanders, Robert J. Kaiser, Peter Hughes, Chris Dodd,
Charles R. Connell*, Cheryl Heiner*, Stephen B. H. Kent & Leroy E. Hood**



Instead of one at a time, Lee Hood et al. labeled each ddNTP with a different fluorophore, allowing for simultaneous imaging

1st Generation Sequencing Era

Polymerase chain reaction - PCR



Wikipedia

PCR invented by Kary Mullis at Cetus – patents filed and first paper in 1985

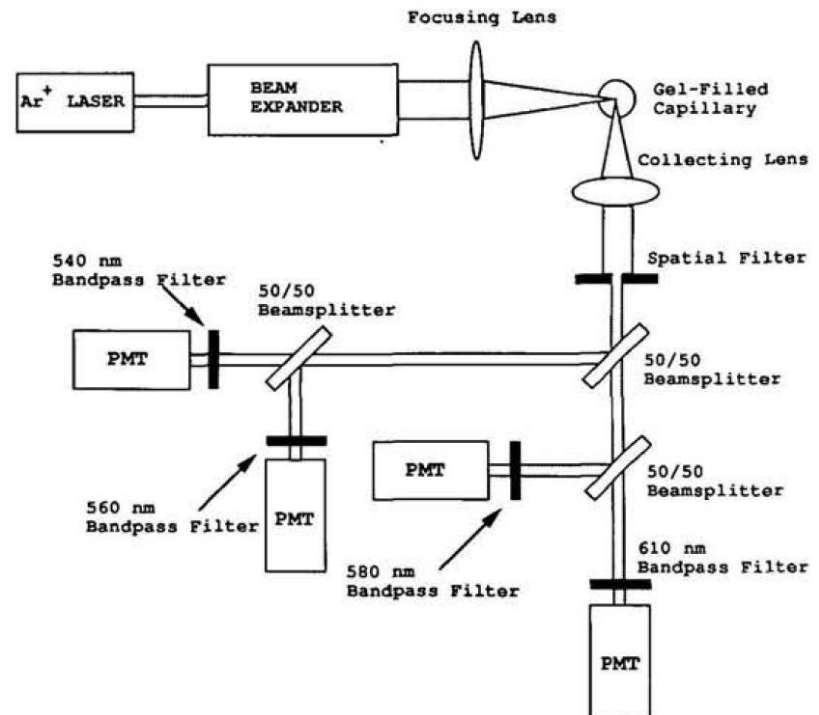
Allows for easy generation of significant amounts of template

1st Generation Sequencing Era

High speed DNA sequencing by capillary electrophoresis

John A. Luckey, Howard Drossman, Anthony J. Kostichka, David A. Mead, Jonathan D'Cunha,
Tracy B. Norris and Lloyd M. Smith*
Department of Chemistry, University of Wisconsin, Madison, WI 53706, USA

Received May 17, 1990; Revised and Accepted June 15, 1990



Applied Biosystems (ABI) the dominant machine/ddNTP provider.

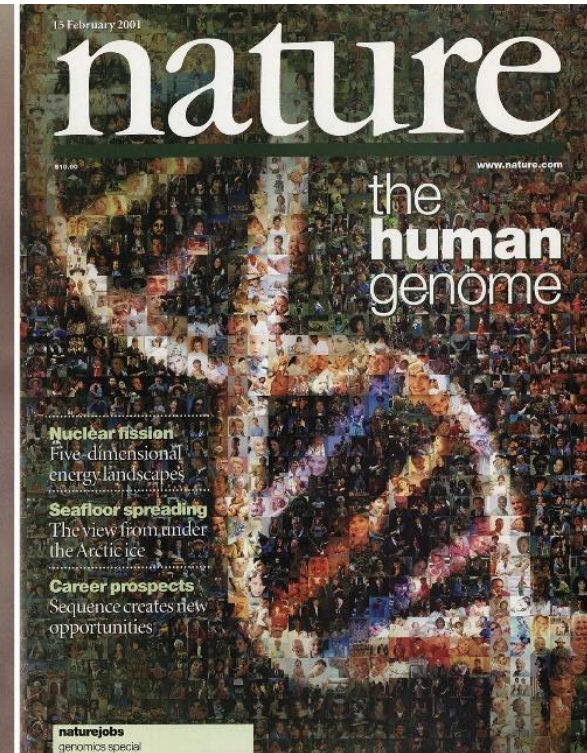
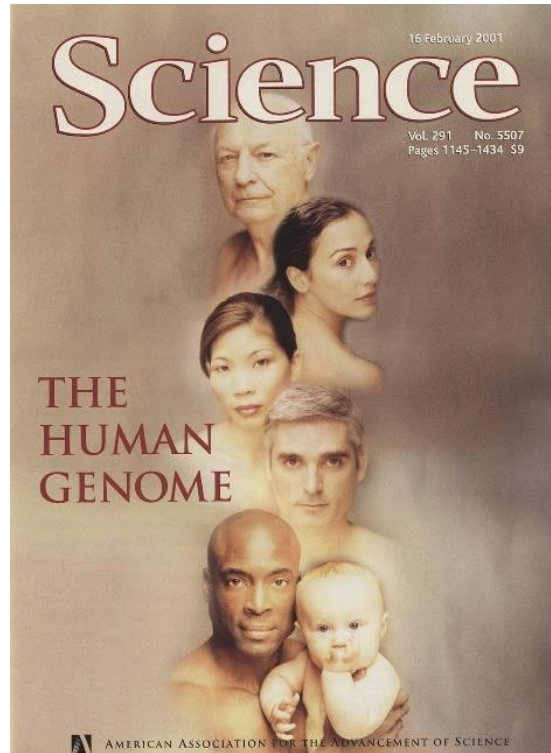
Human Genome Project

Lasted from 1990 to 2003; \$3 billion joint DOE-NIH project.

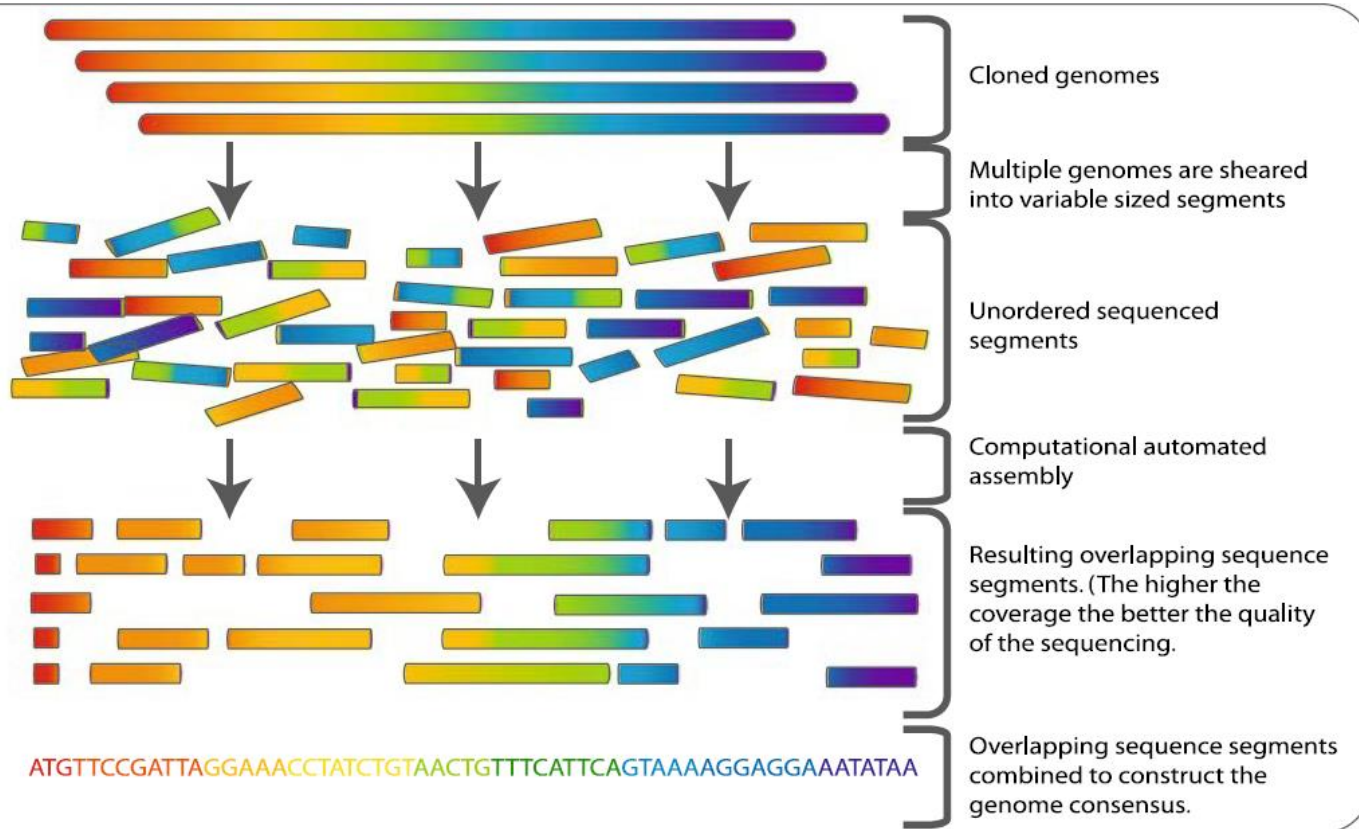
Goal: Sequence the entire human genome in 15 years
First draft done in 2000

Competing Project by Celera Genomics (Craig Venter) for \$300 million

- They were going to patent genes
- President Clinton announced in 2000 genes could not be patented – destroying Celera stock and generally hitting NASDAQ hard
- Both papers came out essentially simultaneously - Nature and Science



Genome Sequencing Methods

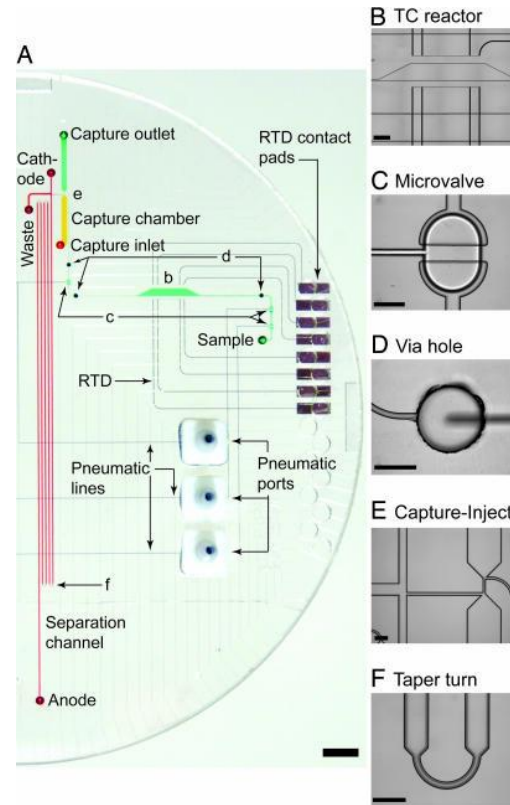
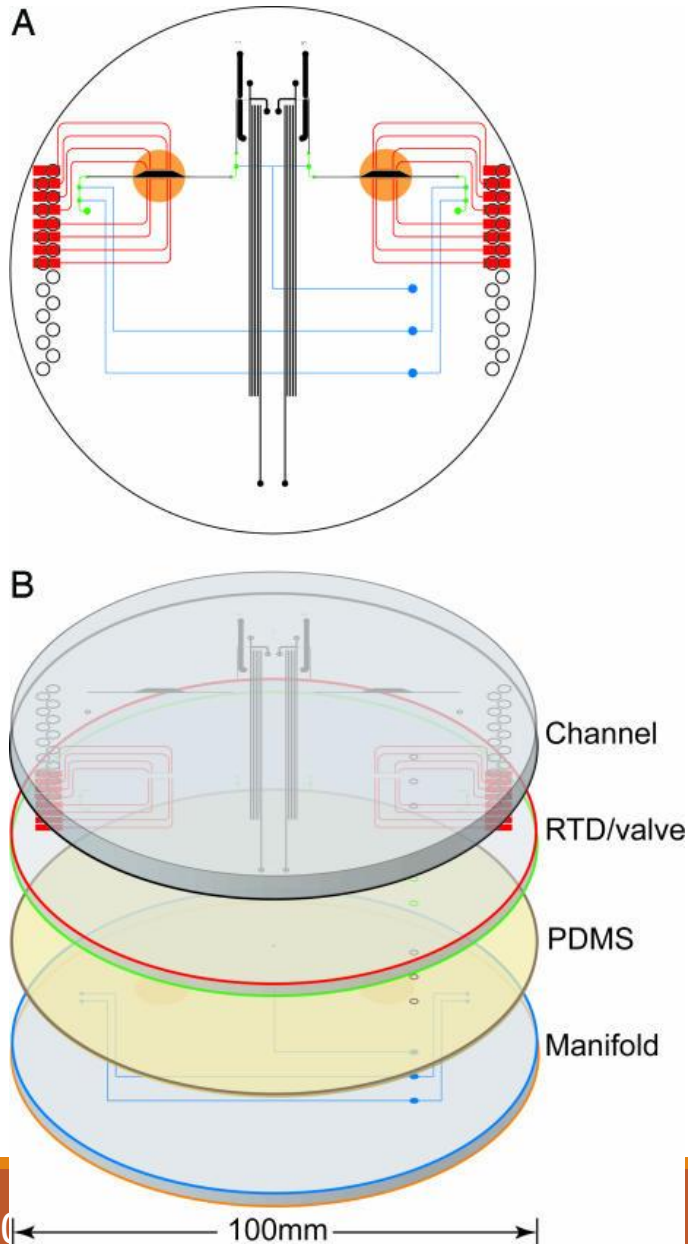


Commins, J., Toft, C., Fares, M. A. Biol. Procedures Online (2009).

Shotgun Sequencing

- Pros: Requires no prior knowledge, fast
- Cons: Random nature makes assembly hard, coverage (number of reads in an area) may be biased

Sanger Sequencing: Microfabrication

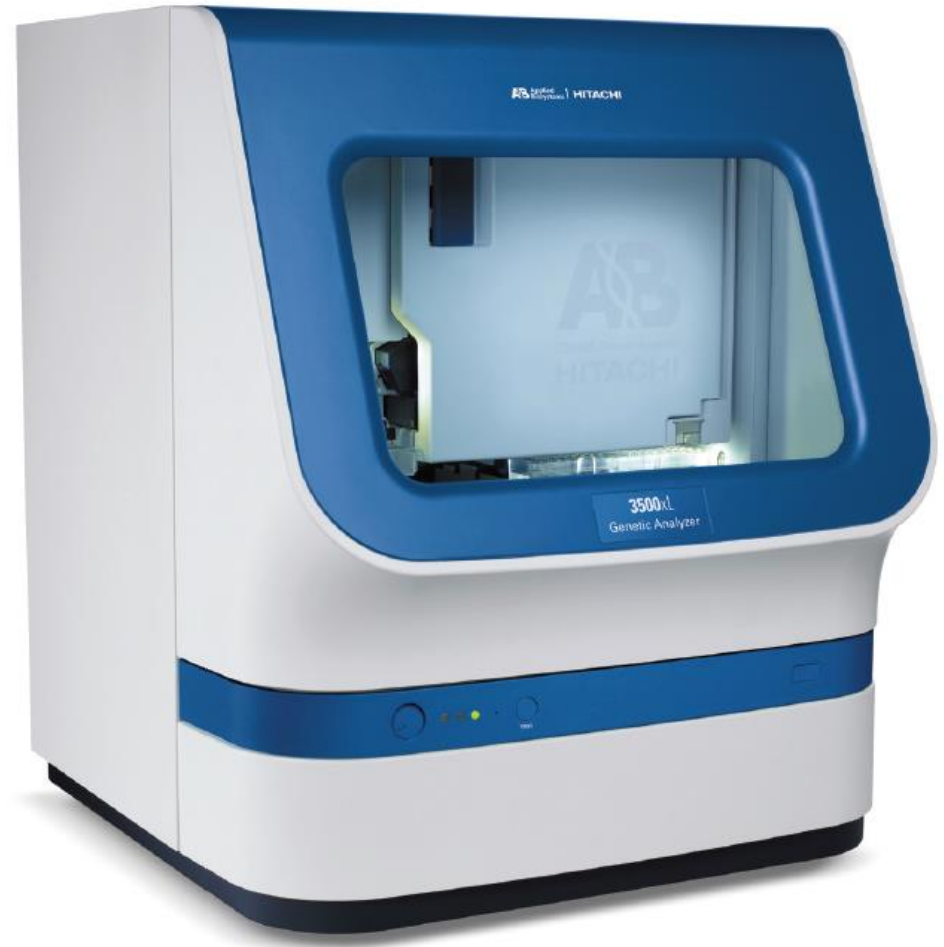


In the mid 2000s several groups, most prominently Mathies et al. made efforts to miniaturize further Sanger sequencing

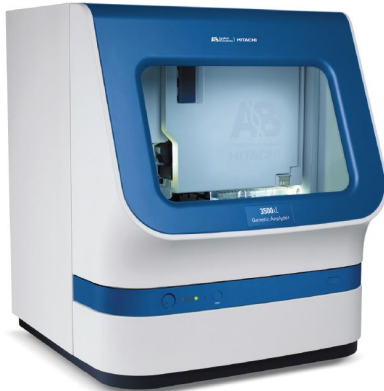
Robert J. Boyle, Pankaj Kumar, and Richard A. Mathies Proc Natl Acad Sci U S A. 2006 May 9; 103(19): 7240–7245.

Sanger Sequencing: End Point

- ABI, since acquired by Life Tech, then by Thermo Fisher, still makes and sells the industry standard capillary electrophoresis machine/sequencer
- Still useful for specific, small scale sequencing tasks
- Certain standard sequencing tasks still use capillary electrophoresis – forensic DNA sequencing, cloning verification.

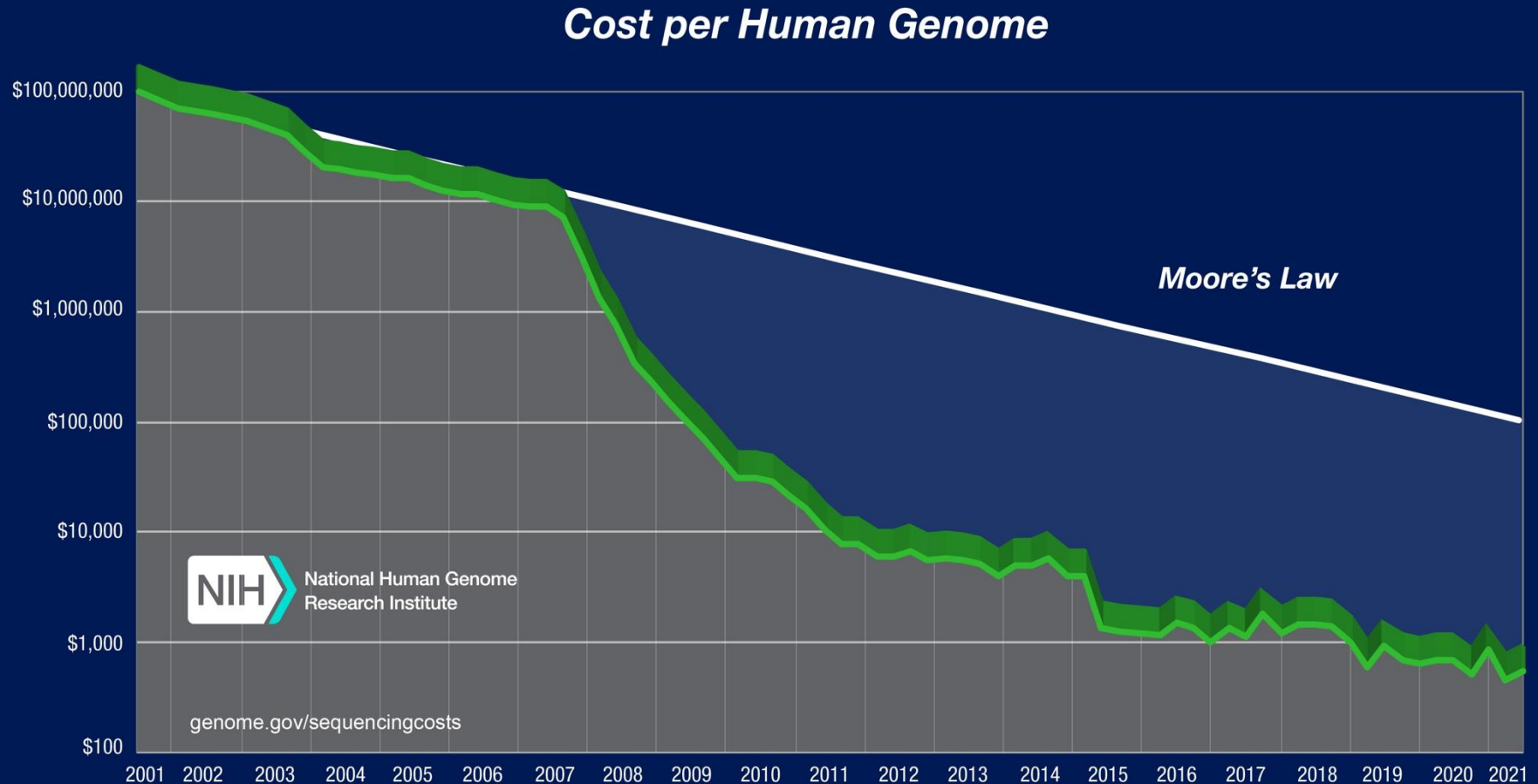


Sequencing: The Next-Generation



In the 2000s “Next-Generation” sequencing is developed – we’re skipping over microarray technology

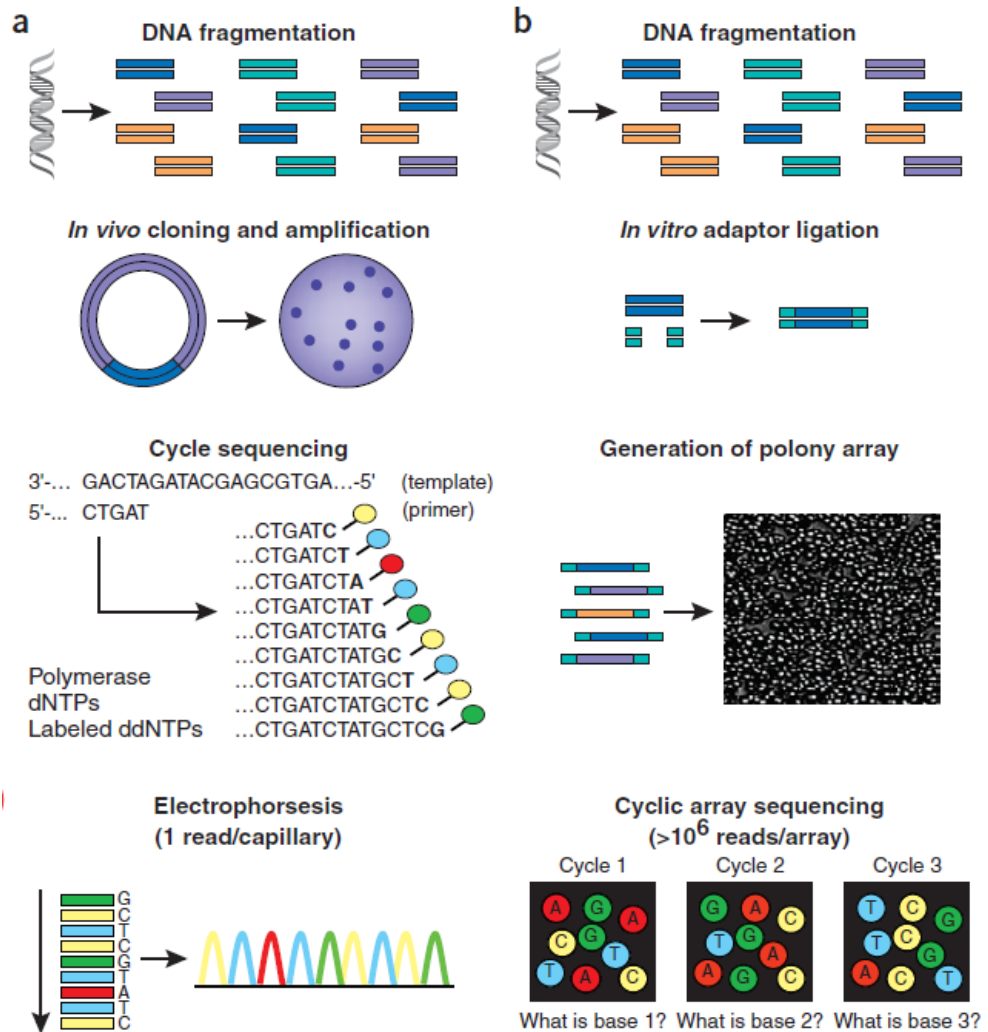
Sequencing Costs



Sequencing costs dropped dramatically with introduction of 2nd gen sequencing.

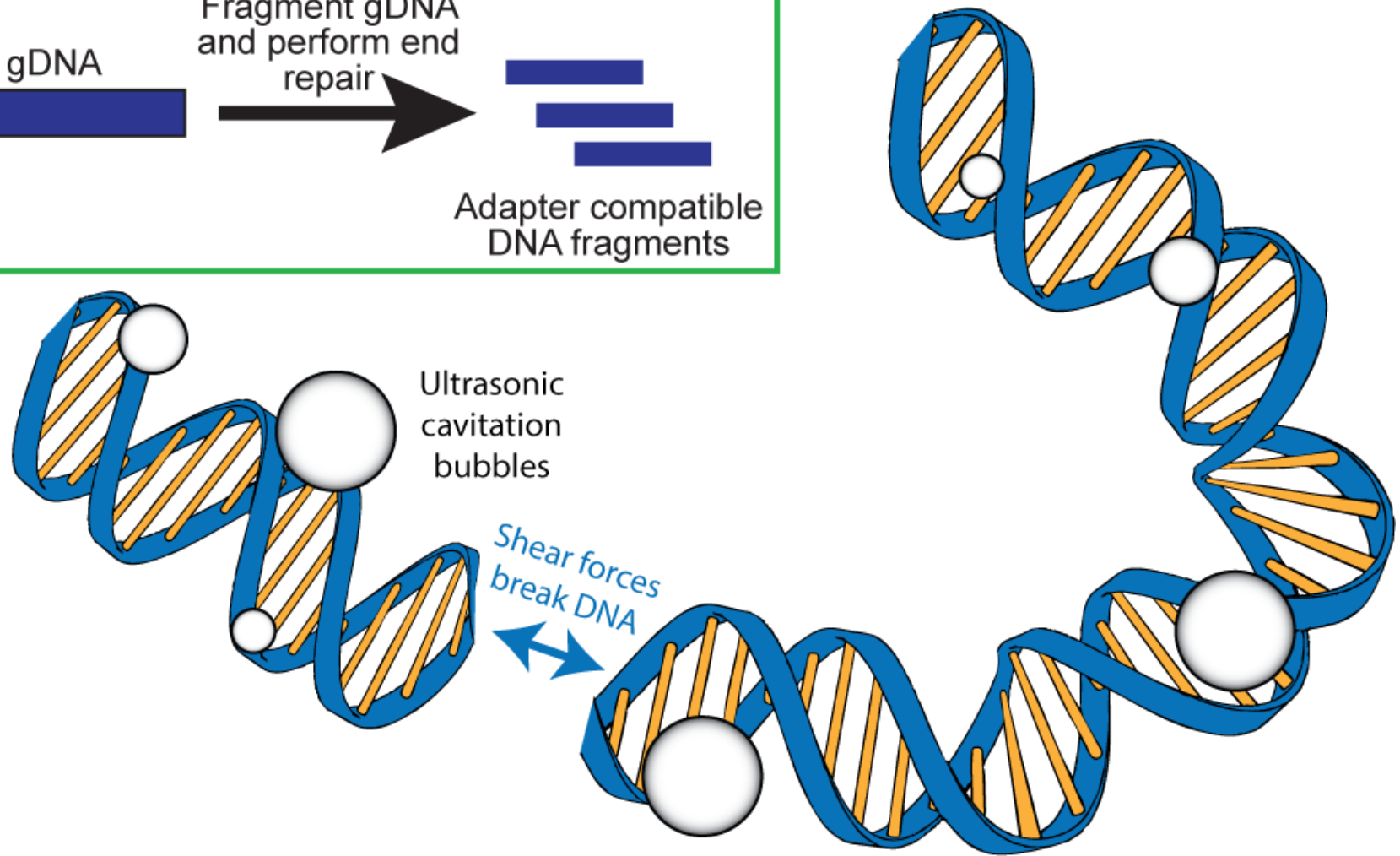
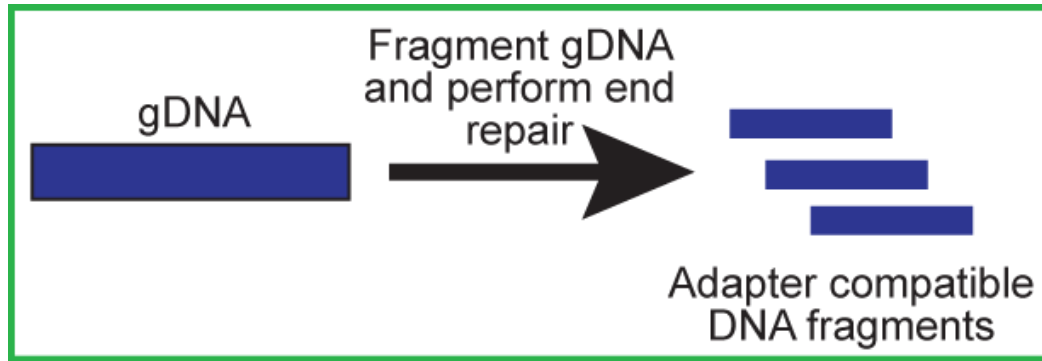
Sequencing: The Next Generation

- Instead of cloning into a plasmid and growing up bacteria, then picking (to isolate one molecule), a combination of adaptor ligation and PCR colony is used
- Instead of size separation, a sequencing-by-synthesis method is used to read the sequence of each polony.



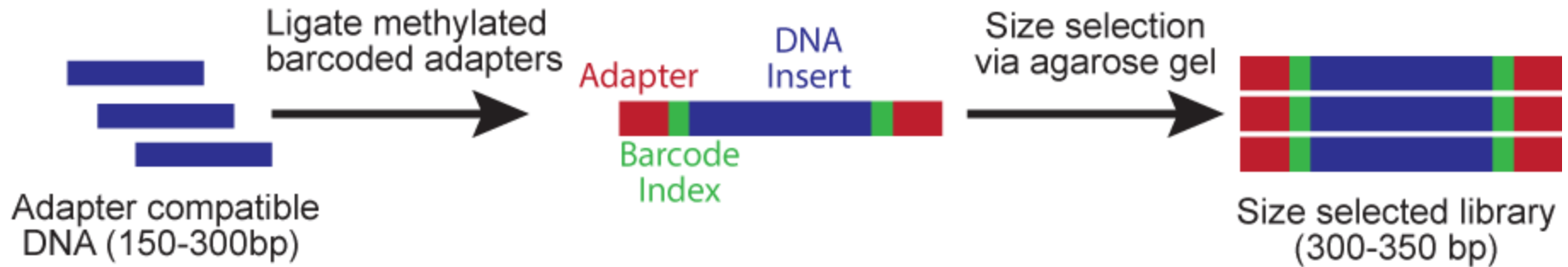
Shendure and Ji Nature Biotechnology (2008)

Next-Generation Library Prep

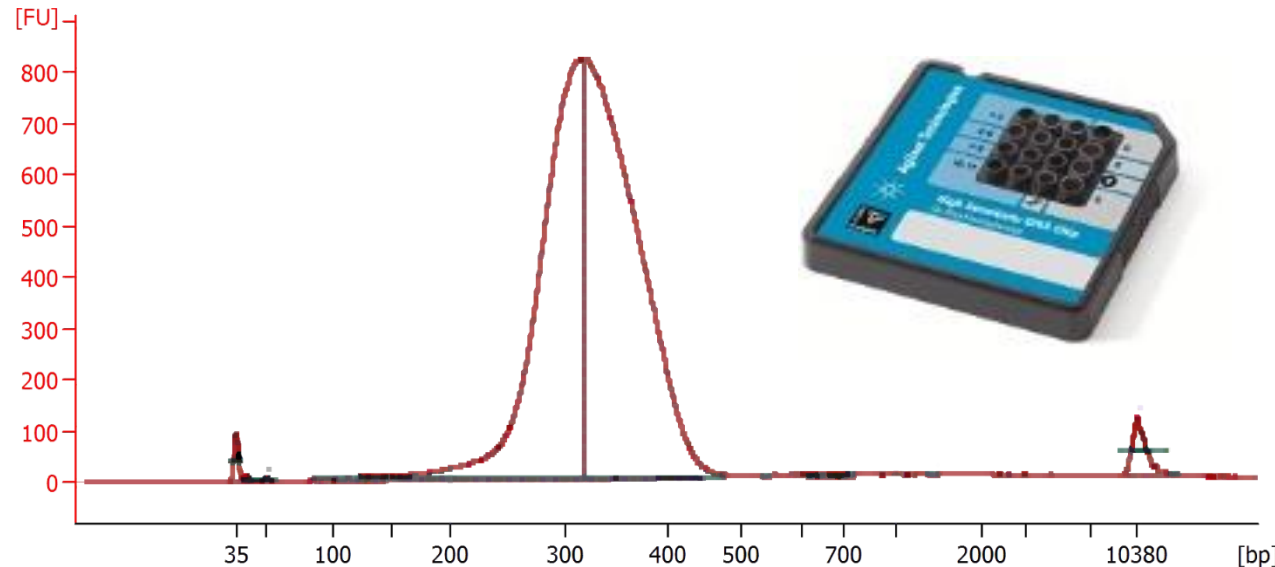


The first step in next-generation sequencing library formation is DNA shearing, usually with ultrasonic cavitation

Ligation/Size Selection



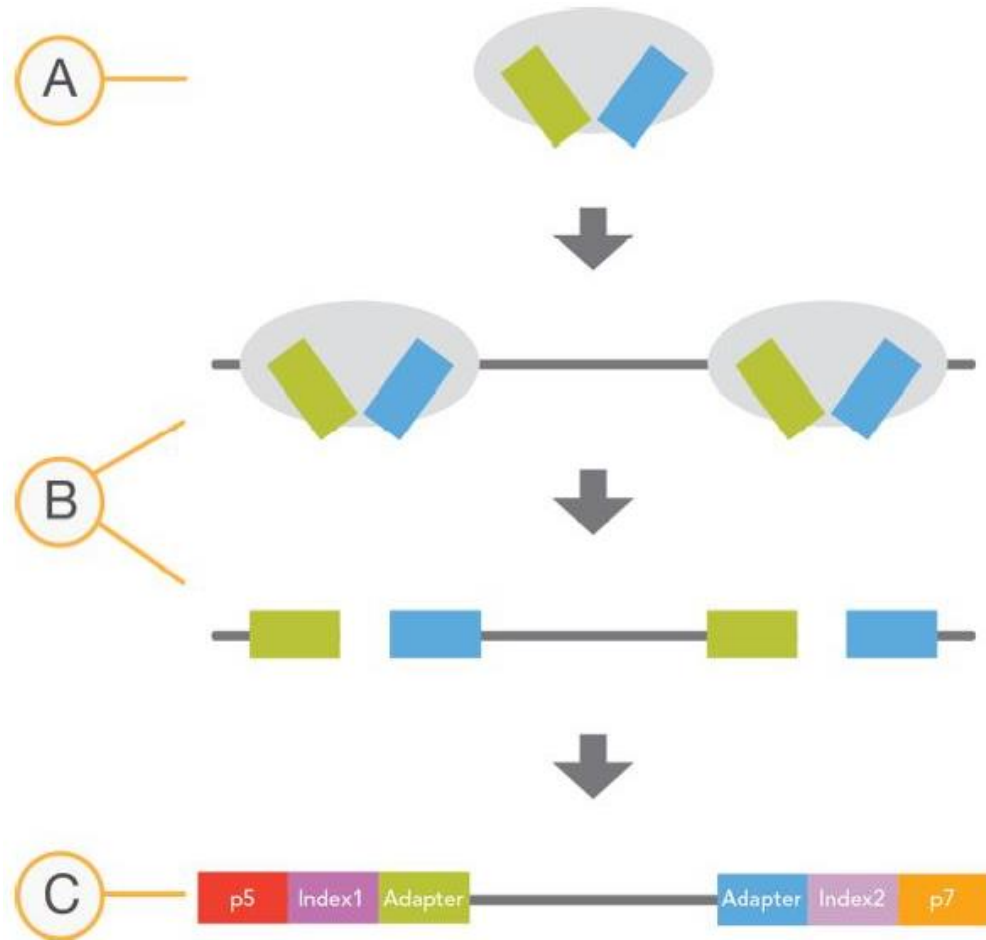
After shearing, platform specific adapters, synthetic DNA sequences both identifying the sample (barcode) and matching sequences on the instrument, are ligated on the sample.



The DNA size is analyzed on a microfluidic device (Agilent Bioanalyzer), then size selected using an agarose gel.

Nextera Adaptors

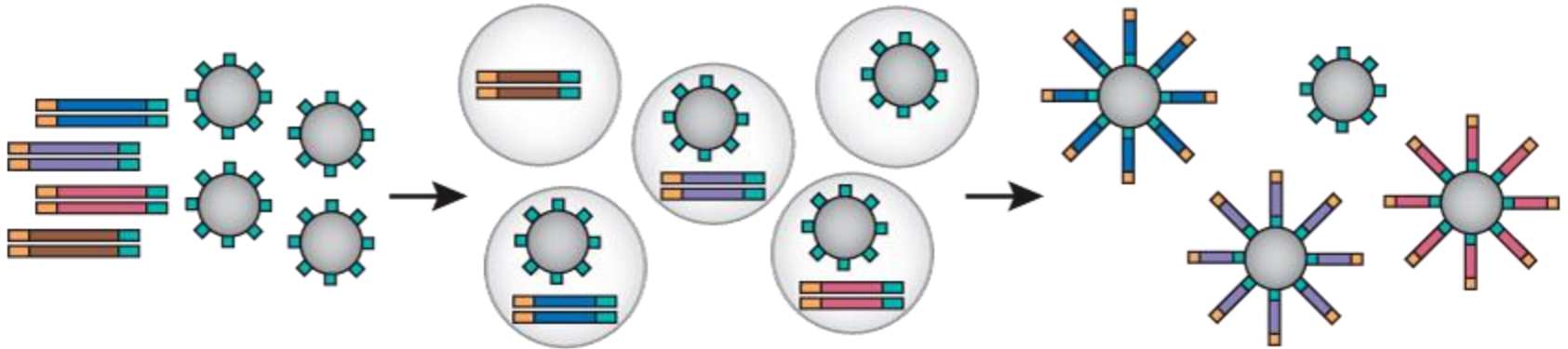
- Alternatively, a transposase based scheme can be used to simultaneously fragment and ligate adaptors.
- Illumina has one such scheme, known as Nextera, using a Tn5 transposase (right).
- MuSeek is another such scheme, from Thermo Fisher.



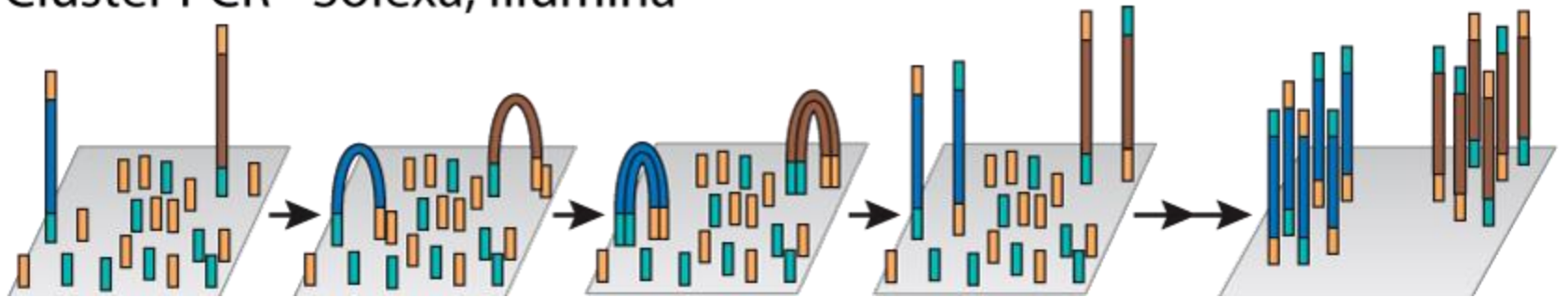
- A Nextera Transposome with Adaptors
B Tagmentation to Fragment and Add Adaptors
C Limited Cycle PCR to Add Sequencing Primer Sequences and Indices

2nd Generation Sequencing

Emulsion PCR - 454, Polonator, SOLiD, Ion Torrent



Cluster PCR - Solexa, Illumina



Shendure and Ji Nature Biotechnology (2008)

PCR Colony, or Polony, first described by Mitra and Church (NAR 1999) takes random genomic DNA fragments, ligates on adaptors, then generates a clonal expansion – a large group of the same molecules in a physically isolated space, either in water in oil droplet (emulsion PCR) or physical location (cluster PCR)

Sequencing with Paired Ends



Reference

This is really the best way to do sequencing

Single-reads

This is

...

is really

...

really the

...

the best

...

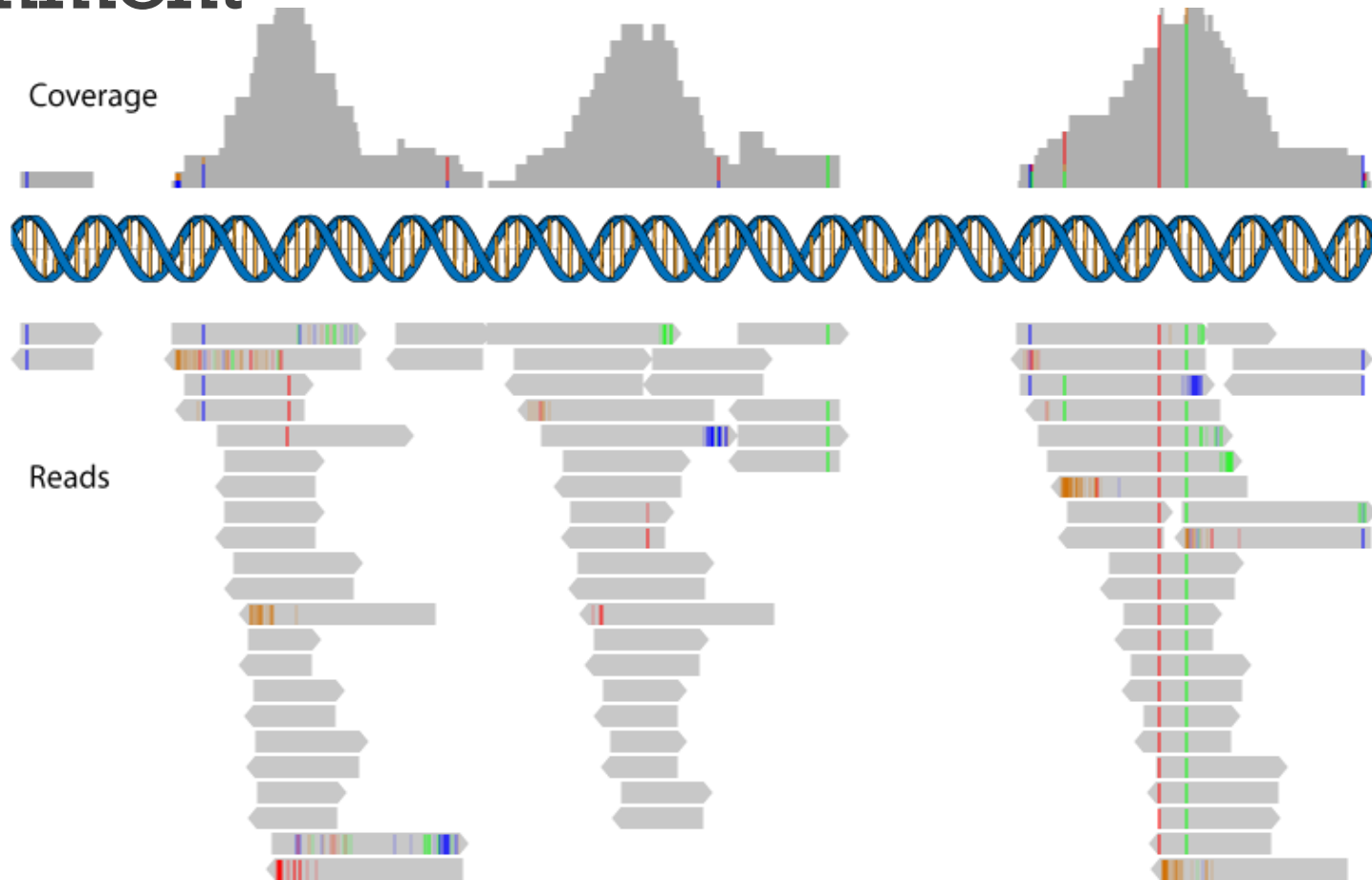
sequencing

Paired-reads

This is (----100 characters-----) sequencing

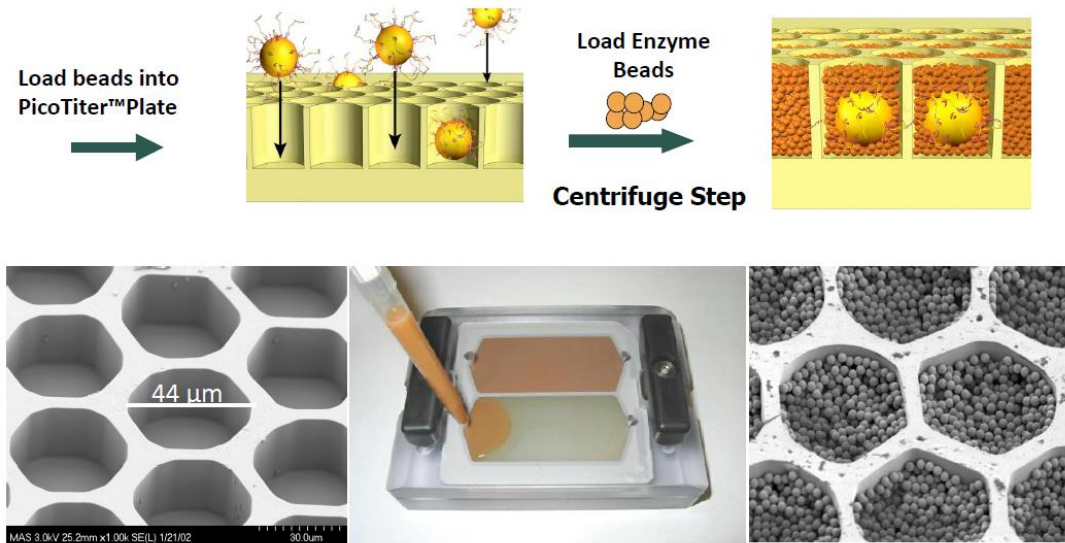
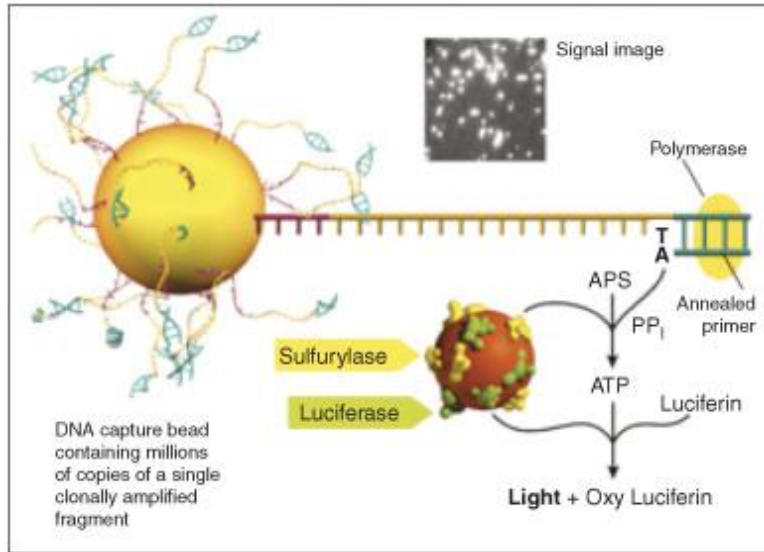
Assembly becomes easier!!

Alignment



- Using a short-read aligner (e.g. bowtie2) we can align these short sequence reads – match up where they belong using a reference genome.
- Measure the difference of these reads versus the reference (colored bands) or the coverage level (peaks)

454 Life Sciences



454 Corp

- When DNA is extended by a base, it releases Ppi (pyrophosphate) and H⁺
- Pyrophosphate is used by luciferase (enzyme) to produce light
- Beads with a polony (from emulsion PCR) are isolated in wells coupled to a fiber optic, then packed in with smaller beads with enzyme/reagents.
- At optimum, gave 1kb long reads, 1M reads, 1Gb yield.

454 Life Sciences



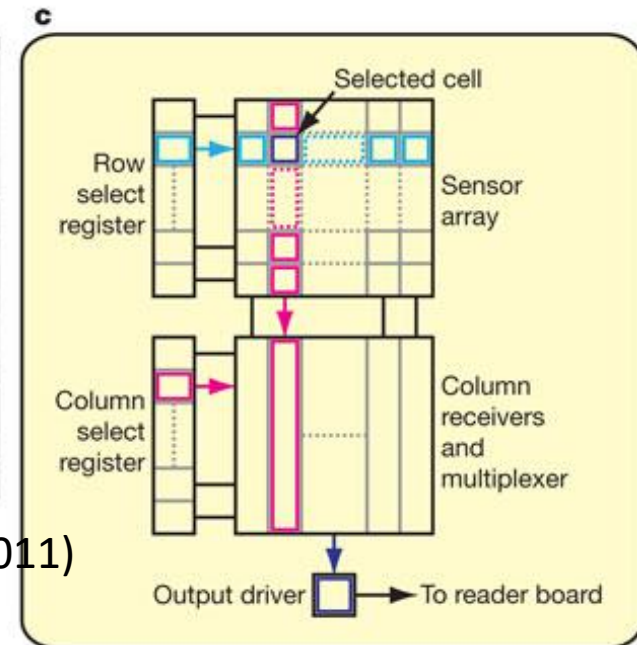
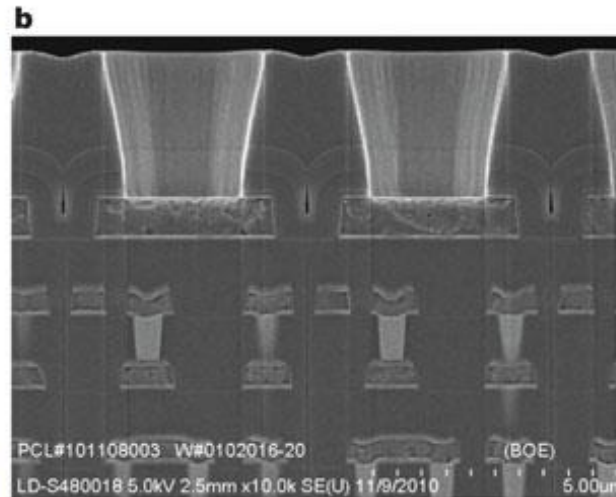
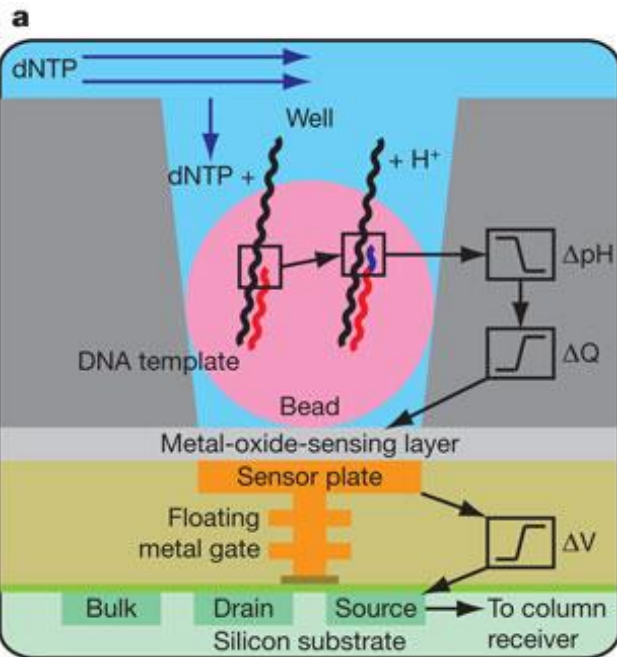
GS20



GS FLX+

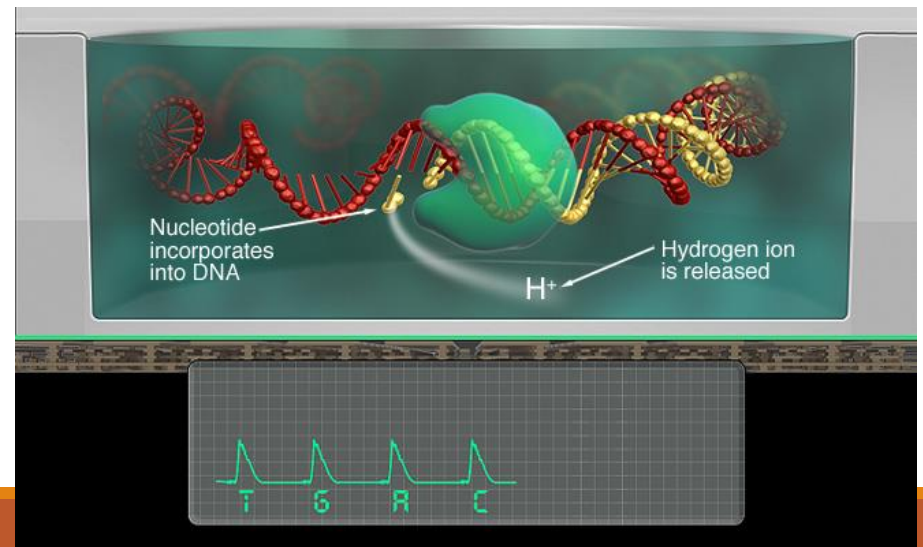
- Founded in 2000 by Rothberg
- First 2nd generation sequencer on the market in 2005 (GS20) – 100bp reads; 20Mb runs
- Bought by Roche in 2007, platform killed 2013, end-of support 2016

Ion Torrent



Rothberg et al. Nature (2011)

- Ion Torrent uses an isFET to measure the number of bases incorporated at one time, sequentially flowing on A, C, G, then T, and repeating.
- Homopolymer runs (e.g., AAAA) will register as peaks of larger area.

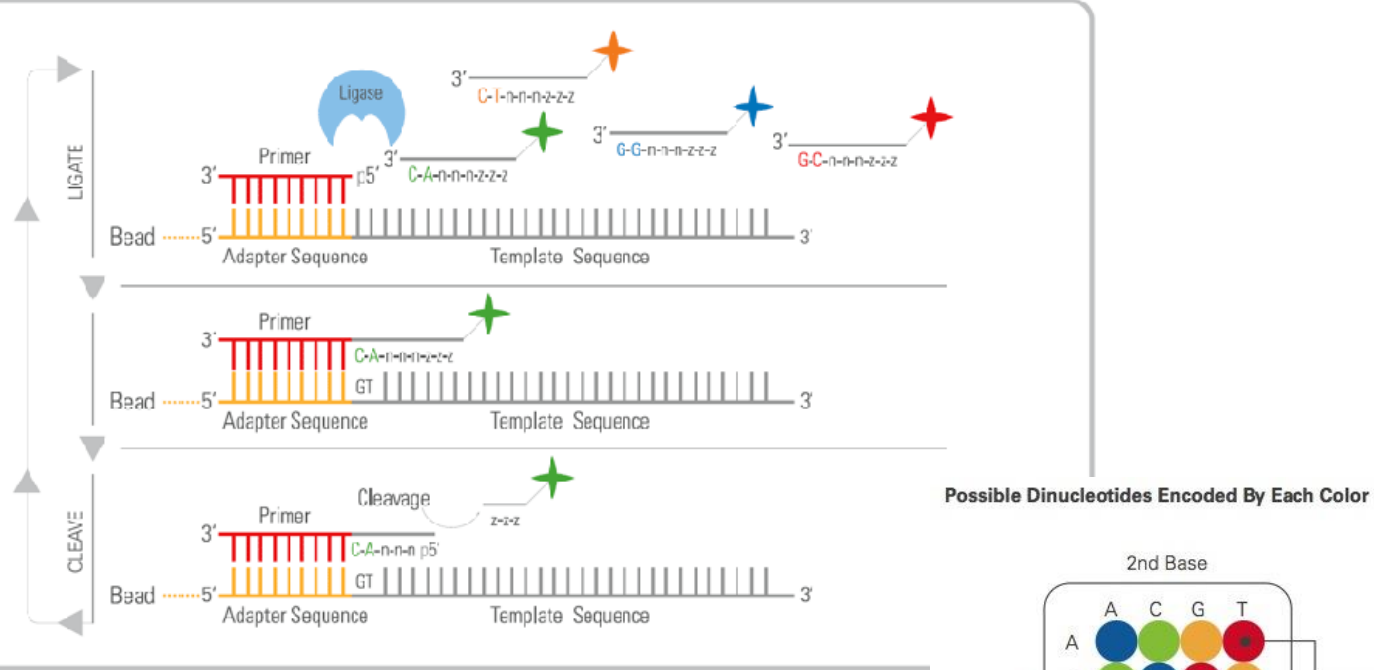


Ion Torrent



- Founded by Rothberg (again) in 2007
- Ship first Personal Genome Machine (PGM) in Dec 2010 – different chips for different yield ranging from 10Mb-1Gb, 200bp read length
- Ion GeneStudio S5 System (60-80M reads/day, 15Gb/day)
- Biggest complaint – library prep is obnoxious, yields don't measure up.

SOLiD Sequencing

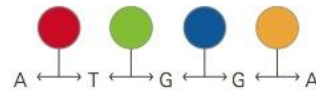


Sequencing by Oligonucleotide Ligation and Detection (SOLiD) - not sequence-by-synthesis, but sequence-by-ligation. Uses emulsion PCR and beads deposited on surface

“Color-space” reads – colors are not unique for all possible 16 bases (dinucleotide match), but each base gets read twice – automatic error checking.

Double Interrogation

With 2 base encoding each base is defined twice

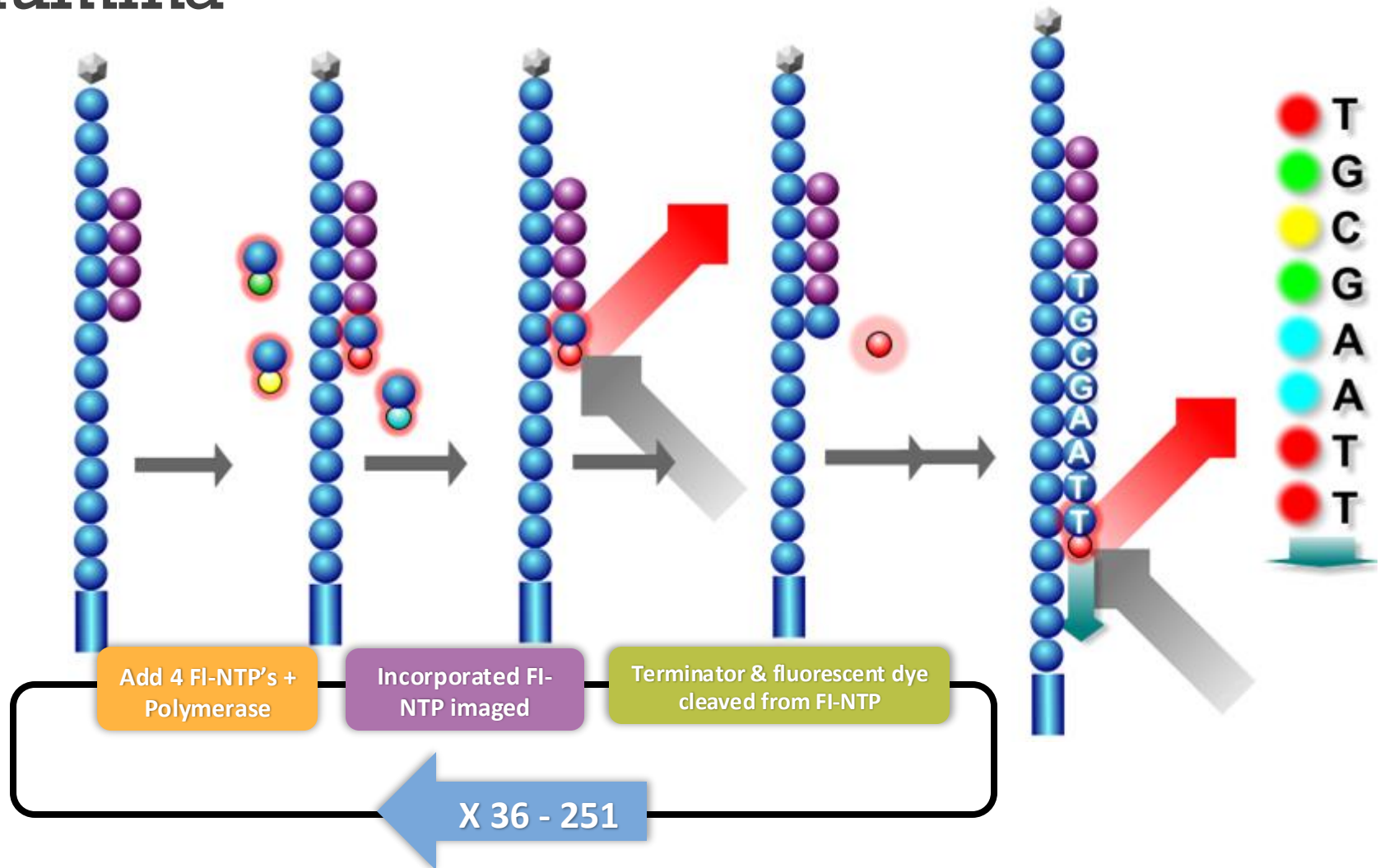


SOLiD Sequencing



- Acquired by ABI from Agencourt (Beckman Genomics)
- First commercial instrument in 2006 (35bp reads, 2-3Gb)
- ABI 5500 released in 2012 (2x50 bp reads, 160Gb)
- Primary limitations/problems: Complicated data compared to competitors (colourspace), small read lengths/data yield limitations.
- Abandoned by ABI/Fisher in favor of Ion Torrent

Illumina



Illumina sequencing uses 4 fluorescently labeled, blocked nucleotides, followed by imaging, and block/fluorescent cleavage to sequence one base at a time.

Illumina



ebay



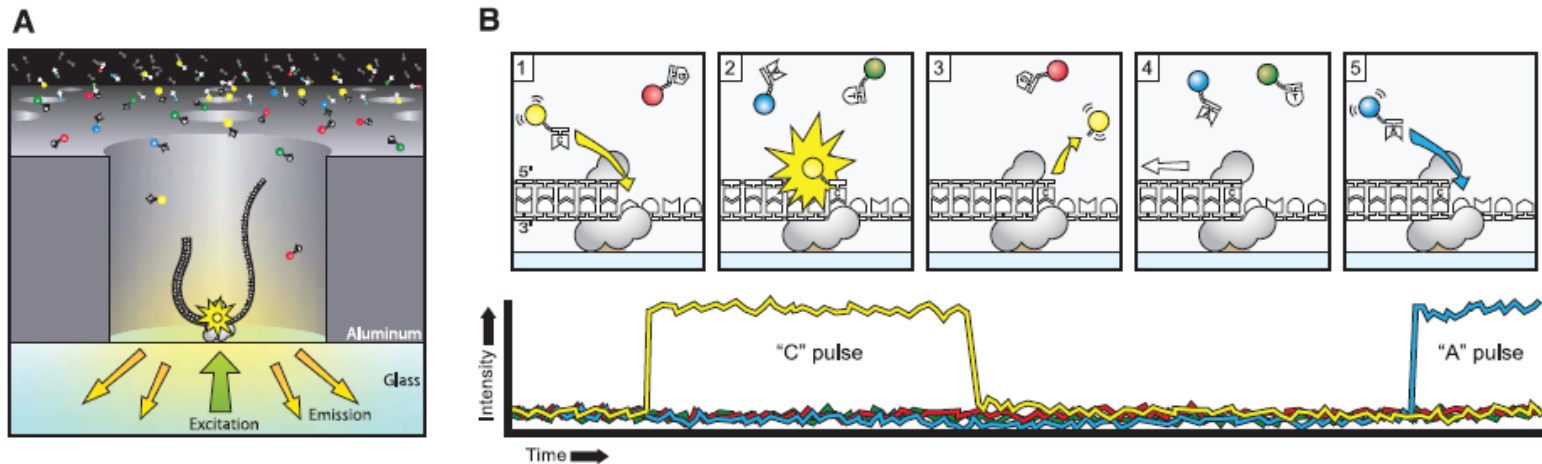
Illumina

- Solexa acquired seq technology from Manteia Predictive Medicine in 2004 to get cluster PCR tech.
- In 2007 Solexa was acquired by Illumina (formerly a DNA microarray company; founded in 1998).
- GA released in 2005, 30bp, 2Gb
- NovaSeq 6000 released in 2017, 2x250, 6Tb (44 hr run)
- Not shown HiSeq 2500, 4000, smaller instruments (iSeq, MiSeq, Miniseq, NextSeq)
- Currently by far the dominant force in sequencing

Pacific Biosystems:

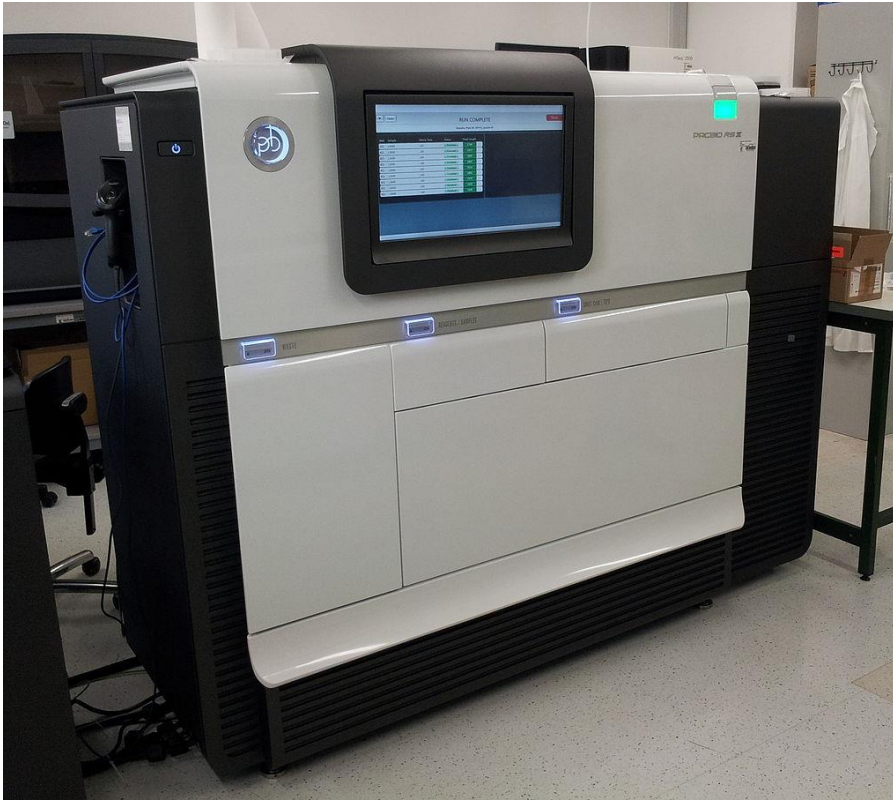
Real-Time DNA Sequencing from Single Polymerase Molecules

SCIENCE VOL 323 2 JANUARY 2009



Long reads (kb long), sensitive to DNA modifications/damage.

Pacbio SMRT seq



Pacific Biosciences founded in 2004 (Craighead et al)

RS released 2011

RSII released 2013 (500Mb)

Sequel released 2016 (partnership with Roche) (10Gb/run)

Sequel II released 2019 – now at about ~60Gb/run ~10-25kb long

Oxford Nanopore

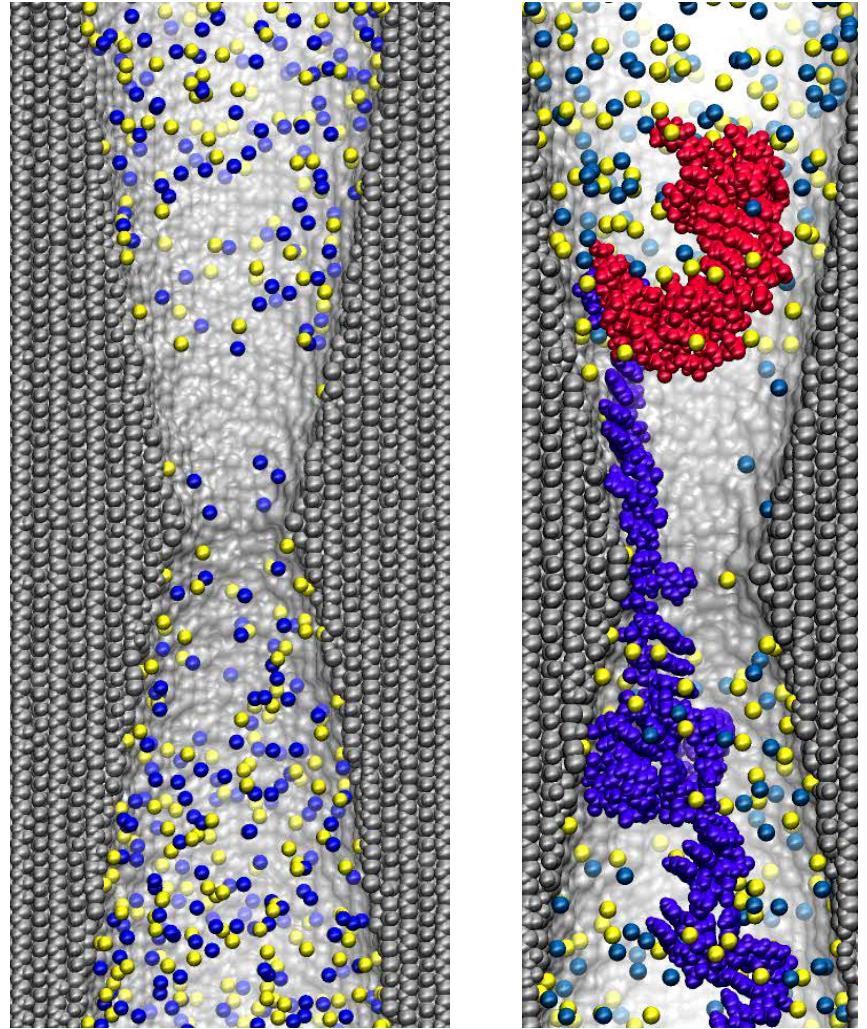


Released in beta in 2014, general public in 2015

3-20Gb per run

Nanopore: Current Blockage

- Basic concept is similar to Coulter counter – detect and characterize objects based on their blockage of a pore
- The pore simply characterizes whatever is placed inside it based on how the molecule occludes current
- We have typically been using it to sequence DNA, but it can be sensitive to any modifications to the DNA molecule, including methylation.



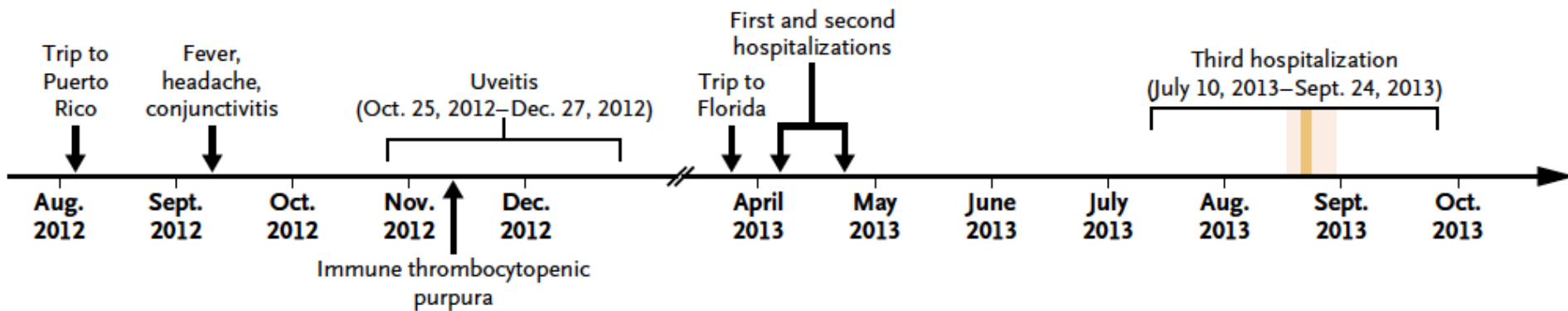
Nanopore operation



Applications: Infectious Disease

Actionable Diagnosis of Neuroleptospirosis by Next-Generation Sequencing

N Engl J Med 2014;370:2408-17.



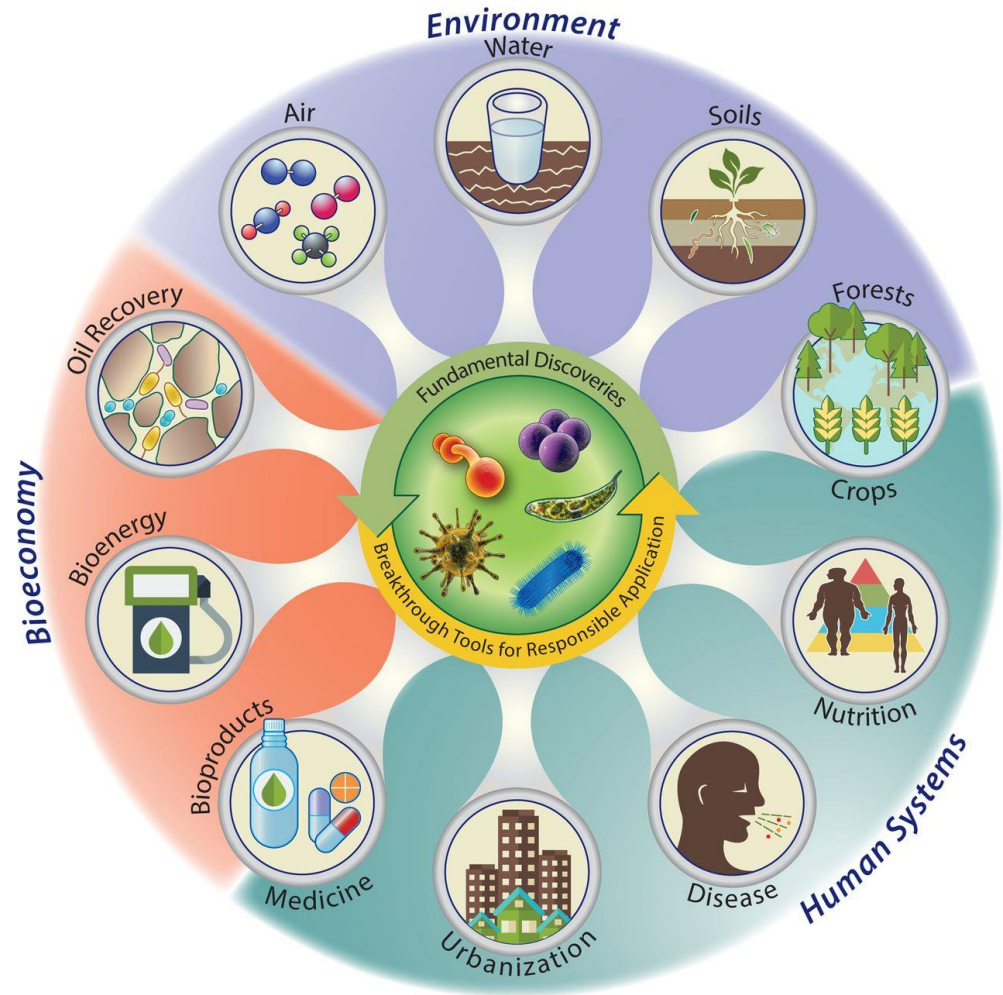
Work from UCSF – 14 year old boy – multiple hospitalizations/infections

Leptospirosis – can't be cultured, hard to see with PCR (too many species)

Treatment was just penicillin G

Exploring the microbiome

- But not just in human samples, bacteria are important in many different types of samples, including:
 - Air
 - Water
 - Soil
 - Plant-microbiome (crops)
 - Synthetic Biology



Blaser et al mBio 2016

Applications: Genomics



#178600

PULMONARY HYPERTENSION, PRIMARY, 1; PPH1

%608995

DYSLEXIA, SUSCEPTIBILITY TO, 8; DYX8

%612596

MULTIPLE SCLEROSIS, SUSCEPTIBILITY TO, 4; MS4

*611409

OCA2 GENE

Databases of (at least) correlation of gene mutations in the human population that coordinate with disease susceptibility or even eye color are known and catalogued.

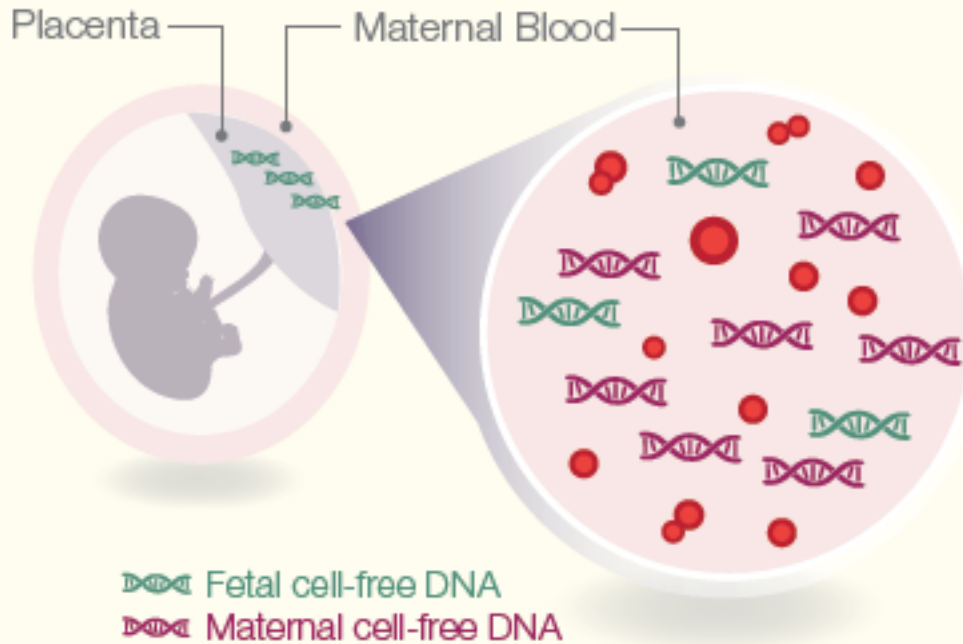
Cancer

Use of Whole-Genome Sequencing to Diagnose a Cryptic Fusion Oncogene

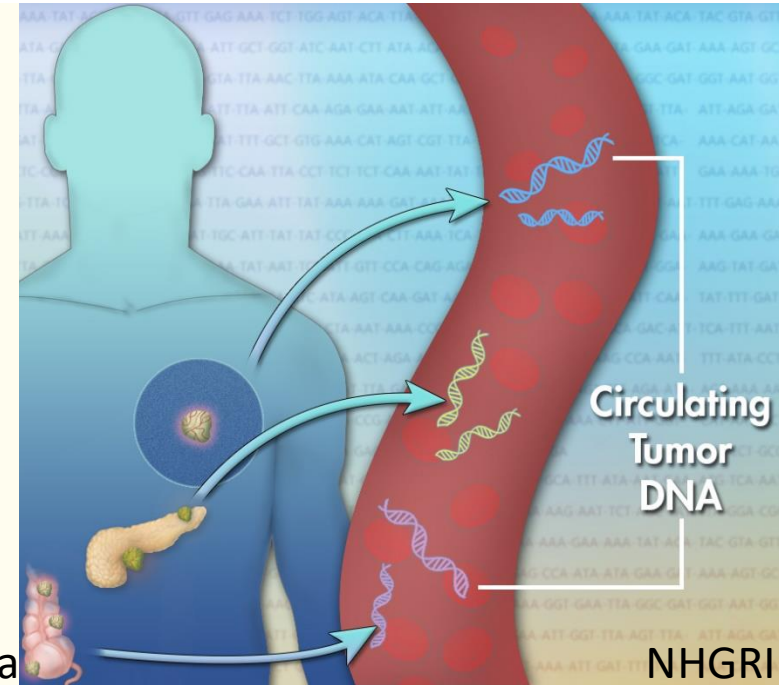
JAMA, April 20, 2011—Vol 305, No. 15

- The timing of chemotherapy drugs in cancer is key
- Often doctors just have to try something for 2 months, then look again
- By the time they can look again, the cancer may have progressed too far
- Better molecular diagnostics provide clear chemotherapeutics
- In this case, they identify a type of Acute Myeloid Leukemia that the patient has is best treated ATRA therapy as compared to allogenic stem cell transplant

DNA screening from blood



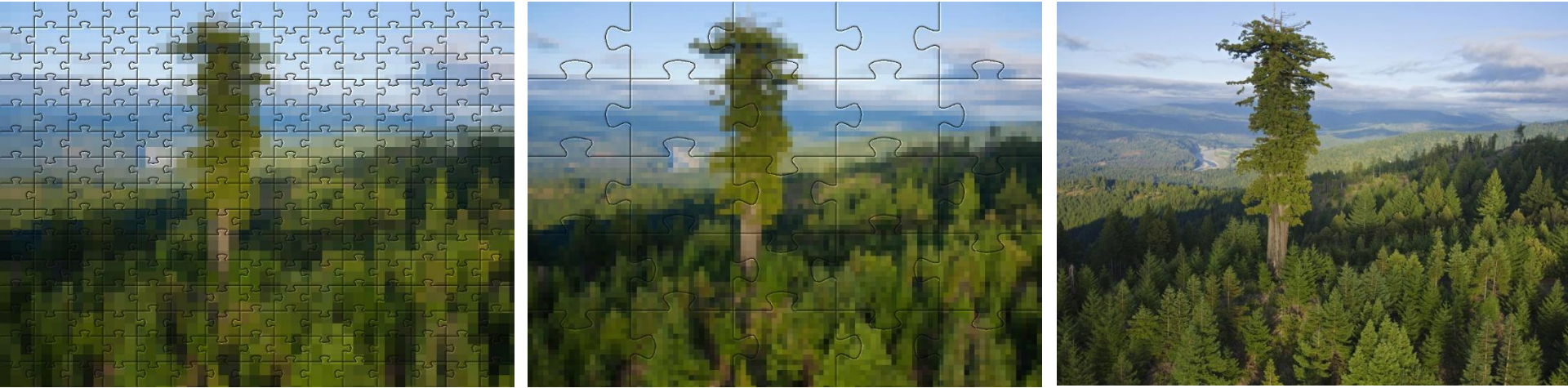
Natera



NHGRI

- By sequencing DNA obtained from blood, different screening testing is available
- Fetal DNA can be sequenced/tested from a pregnant woman's blood, less invasive than amniocentesis, testing for abnormalities
- Tumor cells shed DNA into blood, can be used for screening or surveillance of treatment

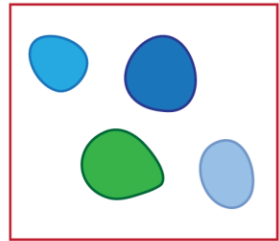
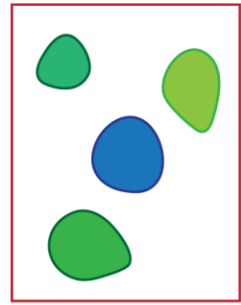
Applications: Building Genomes



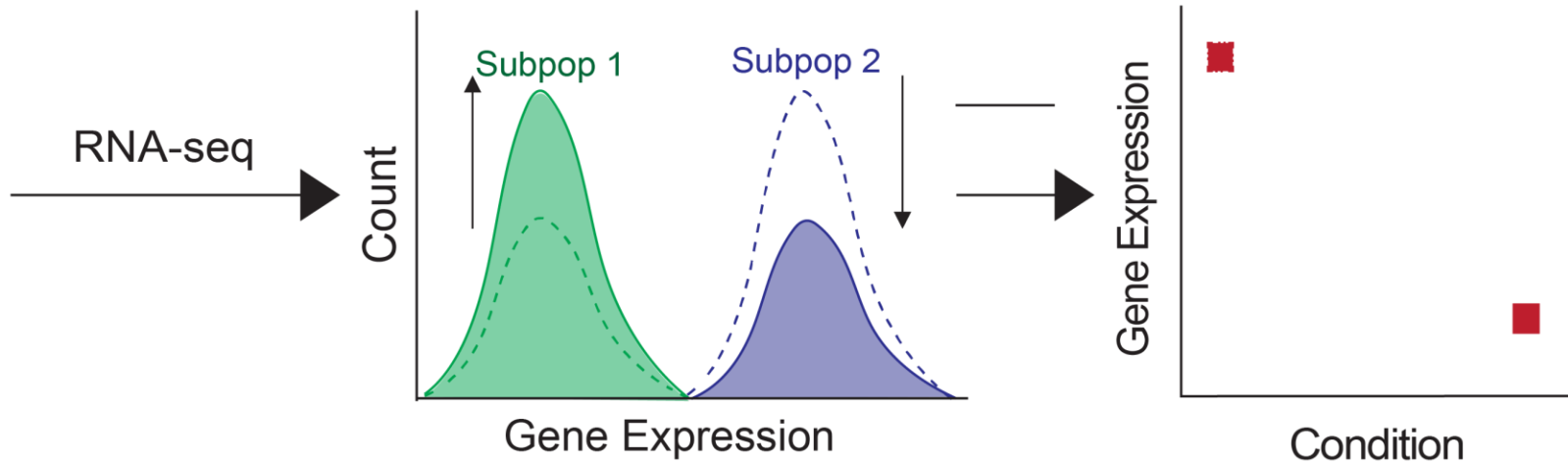
Building a genome is like putting together a puzzle:

- Larger (blurry) pieces = easier puzzle
- Small pieces = hard to put together, can't figure out "blue sky"
- Small + large pieces = clearly and easier to put together

Simpsons paradox?

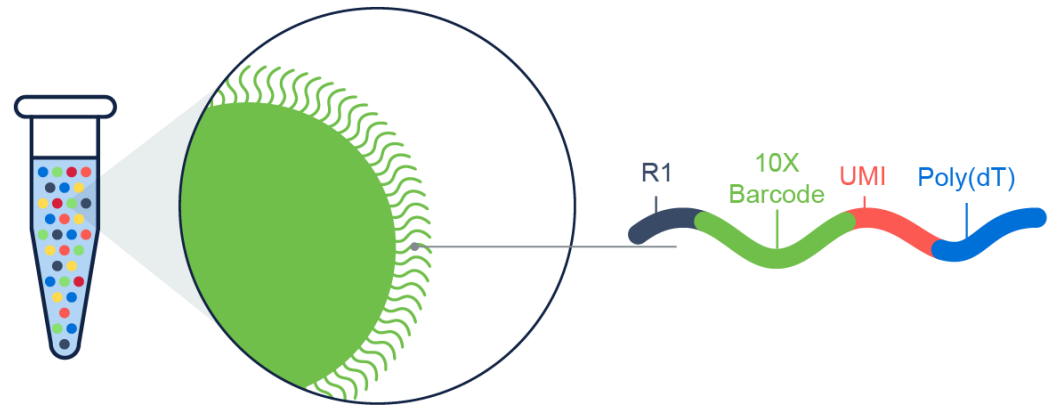
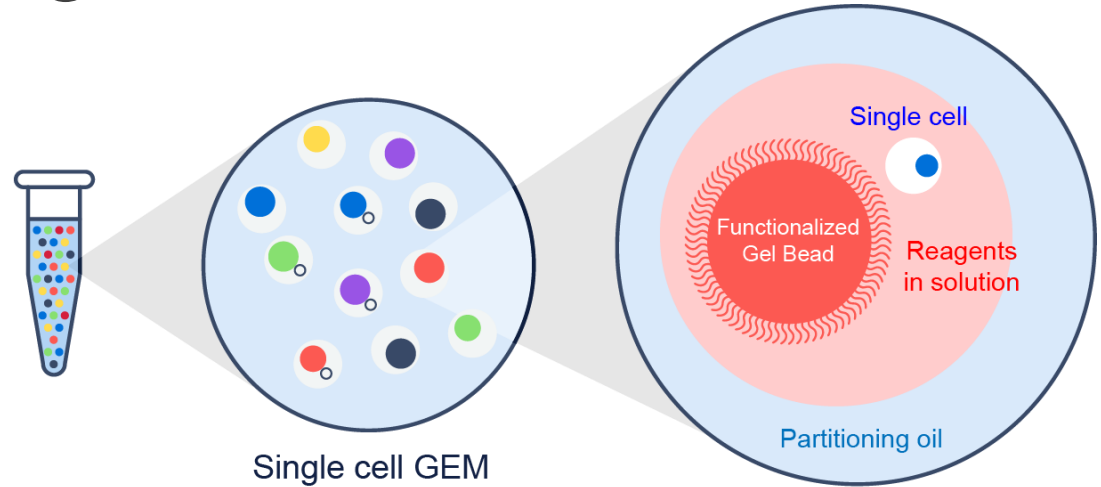
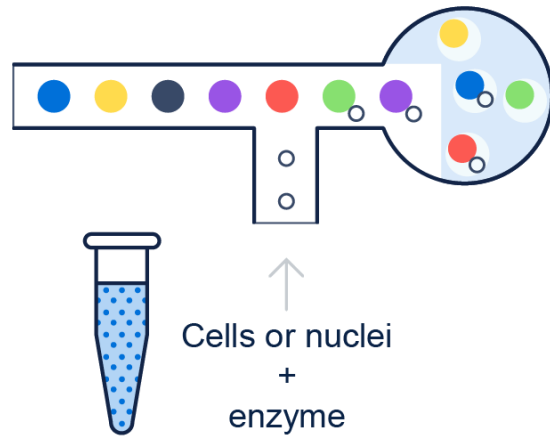


Heterogeneous
Sample



When measuring (for example) gene expression in heterogeneous samples – we can run into Simpson's Paradox, even though the level of gene expression may not change, which cells (or the gene expression of a cell type) might change.

Droplet based single cell



Isolating cells in droplets along with beads with nucleic acid barcodes can label each cell

Sci- or split recombine methods

- Instead of droplets, we can use combinatorial indexing (Cao et al., sci-RNA-seq3)
- Multiple cells receive the first barcode, then you pool and redistribute barcoding again.
- By repeating this step, the space of barcodes for individual cells becomes large enough to distinguish.

